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System overview



1/2	**The SINAMICS converter family**
1/2	Overview
1/3	**Drive selection**
1/3	Overview
1/3	More information
1/4	**SINAMICS S200 servo drive system**
1/4	Overview
1/5	Benefits
1/6	Application
1/7	Design
1/11	Integration
1/13	Configuration
1/15	Technical specifications
1/16	Selection and ordering data
1/16	**SINAMICS S200 servo drive system with SIMOTICS S-1FL2**
1/16	**- Low Inertia servomotors**
1/22	**- Medium Inertia servomotors**
1/28	**- High Inertia servomotors**
1/34	Accessories
1/35	More information
1/35	**SINAMICS S200 starter kit**
1/35	Selection and ordering data
1/36	**Recommended SIMATIC S7 controller**
1/36	Selection and ordering data

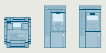
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Overview

SINAMICS frequency converters

SINAMICS frequency converters are the ultimate solution for all drive applications. From low voltage to medium voltage to direct current (DC), our frequency converters meet your needs.

With increased efficiency and versatility, take your drive applications to the next generation for a digital and sustainable future.
www.siemens.com/sinamics

Low voltage											Direct voltage
Standard performance frequency converters		Distributed frequency converters	Industry-specific frequency converters		Servo converters			High performance frequency converters			DC converters
											
SINAMICS V20 G120C G120	SINAMICS G130 G150	SINAMICS G115D G120D SIMATIC ET 200pro FC-2	SINAMICS G120X	SINAMICS G180	SINAMICS V90 S200	SINAMICS S110	SINAMICS S210 (6SL5...)	SINAMICS G220	SINAMICS S120 S120M	SINAMICS S150	SINAMICS DCM DCP [1]
0.12 kW to 250 kW	75 kW to 2700 kW	0.37 kW to 7.5 kW	0.75 kW to 630 kW	2.2 kW to 6600 kW	0.05 kW to 7 kW	0.55 kW to 132 kW	0.05 kW to 7 kW	0.55 kW to 110 kW	0.55 kW to 5700 kW	75 kW to 1200 kW	6 kW to 30 MW
Pumps, fans, compressors, conveyor belts, mixers, mills, spinning machines, textile machines, refrigerated display counters, fitness equipment, ventilation systems, single-axis positioning applications in machine and plant engineering	Pumps, fans, compressors, conveyor belts, mixers, mills, extruders	Conveyor technology, single-axis positioning applications (G120D)	Pumps, fans, compressors, building management systems, process industry, HVAC, water/waste water industries	Pumps, fans, compressors, conveyor belts, extruders, mixers, mills, kneaders, centrifuges, separators	Handling machines, packaging machines, automatic assembly machines, metal forming machines, printing machines, winding and unwinding units	Single-axis positioning applications in machine and plant engineering	Packaging machines, handling equipment, feed and withdrawal devices, stacking units, automatic assembly machines, laboratory automation, wood, glass and ceramics industry, digital printing machines	Pumps, fans, compressors, conveyor belts, mixers, mills, spinning machines, textile machines, refrigerated display counters, fitness equipment, ventilation systems, single-axis positioning applications in machine and plant engineering	Production machines (packaging, textile and printing machines, paper machines, plastic processing machines), machine tools, plants, process lines and rolling mills, marine drives, test bays	Test bays, cross cutters, centrifuges	Rolling mill drives, wire-drawing machines, extruders and kneaders, cableways and lifts, test bay drives
Catalog D 31.1	Catalog D 11	Catalog D 31.2	Catalog D 31.5	Catalog D 18.1	Catalog D 33 D 37.1	Catalog D 31.1	Catalog D 32	Catalog D 36.1	Catalogs D 21.3, D 21.4 NC 62	Catalog D 21.3	Catalog D 23.1, SiePortal
Engineering tools (e.g. Siemens Product Configurator, TIA Selection Tool, DriveSim Designer/Engineer, STARTER and SINAMICS Startdrive)											

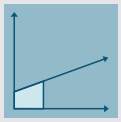
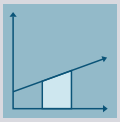
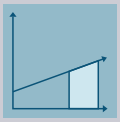
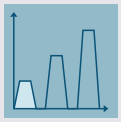
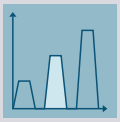
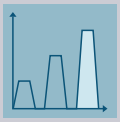
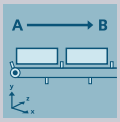
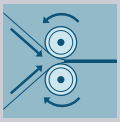
[1] DC/DC controllers, see SiePortal.

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Overview

SINAMICS selection guide – typical applications

Use	Requirements for torque accuracy/speed accuracy/position accuracy/coordination of axes/functionality					
	Continuous motion			Non-continuous motion		
	Basic	Medium	High	Basic	Medium	High
						
Pumping, ventilating, compressing 	Centrifugal pumps Radial / axial fans Compressors	Centrifugal pumps Radial / axial fans Compressors	Eccentric screw pumps	Hydraulic pumps Metering pumps	Hydraulic pumps Metering pumps	Descaling pumps Hydraulic pumps
	V20 G120C G120X	G120X G130/G150 G180 [1] DCM	G220 S120	G120/G220	S110	S120
Moving 	Conveyor belts Roller conveyors Chain conveyors	Conveyor belts Roller conveyors Chain conveyors Lifting/lowering devices Elevators Escalators/moving walkways Indoor cranes Marine drives Cable railways	Elevators Container cranes Mining hoists Excavators for open-cast mining Test bays	Acceleration conveyors Storage and retrieval machines	Acceleration conveyors Storage and retrieval machines Cross cutters Reel changers	Storage and retrieval machines Robotics Pick & place Rotary indexing tables Cross cutters Roll feeds Engagers/ disengagers
	V20 G115D G120C ET 200pro FC-2 [2]	G120/G220 G120D G130/G150 G180 [1]	G220 S120 S150 DCM	V90 **S200** G120/G220 G120D	S110 S210 DCM	S120 S210 DCM
Processing 	Mills Mixers Kneaders Crushers Agitators Centrifuges	Mills Mixers Kneaders Crushers Agitators Centrifuges Extruders Rotary furnaces	Extruders Winders/unwinders Lead/follower drives Calenders Main press drives Printing machines	Tubular bagging machines Single-axis motion control such as • Position profiles • Path profiles	Tubular bagging machines Single-axis motion control such as • Position profiles • Path profiles	Servo presses Rolling mill drives Multi-axis motion control such as • Multi-axis positioning • Cams • Interpolations
	V20 G120C	G120 G130/G150 G180 [1]	G220 S120 S150 DCM	V90 **S200** G120/G220	S110 S210	S120 S210 DCM
Machining 	Main drives for • Turning • Milling • Drilling	Main drives for • Drilling • Sawing	Main drives for • Turning • Milling • Drilling • Gear cutting • Grinding	Axis drives for • Turning • Milling • Drilling	Axis drives for • Drilling • Sawing	Axis drives for • Turning • Milling • Drilling • Lasering • Gear cutting • Grinding • Nibbling and punching
	S110	S110 S120	S120	S110	S110 S120	S120

Using the SINAMICS selection guide

The varying range of demands on modern variable frequency drives requires a large number of different types. Selecting the optimum drive has become a significantly more complex process. The application matrix shown simplifies this selection process considerably, by suggesting the ideal SINAMICS drive for examples of typical applications and requirements.

- The application type is selected from the vertical column
  - Pumping, ventilating, compressing
  - Moving
  - Processing
  - Machining
- The quality of the motion type is selected from the horizontal row
  - Basic
  - Medium
  - High

More information

Further information about SINAMICS is available on the internet at www.siemens.com/sinamics

Practical application examples and descriptions are available on the internet at www.siemens.com/sinamics-applications

[1] Industry-specific converters.

[2] Information on the SIMATIC ET 200pro FC-2 frequency converter is available in Catalog D 31.2 and at www.siemens.com/et200pro-fc

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Overview

SINAMICS S200 servo drive system



Example: SINAMICS S200 servo drive system consisting of a SINAMICS S200 servo converter, 200 ... 240 V 1 AC / 3 AC, PROFINET, frame size FSB, a SIMOTICS S-1FL2 servomotor, shaft height 30, metal hybrid connector, outlet direction drive end (DE) and MOTION-CONNECT 350 dual cables for power and communication

The SINAMICS S200 servo drive system is a single axis AC/AC servo drive system with compact design, high dynamic performance, ease of use and high reliability for motion control applications in standard servo motion market. It comprises SINAMICS S200 servo converters, SIMOTICS S-1FL2 servomotors, MOTION-CONNECT 350 and 380 connection systems and engineering tools.

System components

SINAMICS S200 servo converters

The SINAMICS S200 servo converters consist of a basic version and a standard version with pulse train input position control (PTI) or PROFINET RT/IRT (PN) to meet various needs of applications and work perfectly together with SIMATIC controllers. With the SINAMICS S200, the PROFINET function "Dynamic Frame Packing" (DFP) is available, which enables the connection of large numbers of axes to a SIMATIC controller.

- SINAMICS S200 Basic servo converters, power range 0.1 to 1 kW at 200 ... 240 V 1 AC / 3 AC
- SINAMICS S200 servo converters, power range 0.1 to 1 kW at 200 ... 240 V 1 AC / 3 AC and 0.2 to 7 kW at 380 ... 480 V 3 AC

SIMOTICS S-1FL2 servomotors

With flexible encoder options, including absolute encoders 17-bit singleturn, 21-bit singleturn and 21-bit singleturn + 12-bit multiturn, the system provides precise and accurate performance. The encoders can be used without a battery, making it easy to operate and maintenance.

- Low Inertia for high dynamic performance
- Medium Inertia for dynamic and smooth operational performance
- High Inertia for smooth operational performance

MOTION-CONNECT 350 and 380 connection systems

The cables are available in lengths of up to 30 m and one step further optimized in easy and secure installation with "one-push-lock" motor connecting solution for motor shaft heights 20, 30 and 40.

- MOTION-CONNECT 350 cables, which provide a connection solution for fixed installation applications
- MOTION-CONNECT 380 cables, which provide a connection solution for drag chain installation applications that require a high number of bending cycles

SINAMICS Drive Software highlights from firmware V6.4

- OPC UA server with information model for inventory management and process data acquisition (from firmware V6.4 HF2)
- Security:
  - Security event log for sending security events via syslog protocol to a server
  - Additional UMAC privileges for security configuration
  - Integrity protection for backups
  - Extended handling of certificates
- Integrated Positioning
  - Higher resolution through 64-bit data type for position values together with new telegram 113
  - Load gearbox position tracking support for reproducibility of the load position when using gears
- Vibration suppression support for the reduction of vibrations in the natural frequency range up to 10 Hz
- Configurable supplementary telegram for handover additional data to the PLC

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Benefits

Performance-optimized

Dynamic

Ideal for dynamic applications thanks to:

- fast current controllers
- 8 kHz PWM frequency

Fast

The system enables seamless communication to controllers via PROFINET IRT, providing fast and efficient data exchange and control.

Overload capacity

SIMOTICS S-1FL2 servomotors providing up to 350 % overload capacity, can handle sudden increases in load without compromising performance or efficiency, ensuring uninterrupted operation and increased productivity.

Compact

To enable more efficient use of cabinet space and improve application system design, both servo converters and servomotors are further optimized in compactness thanks to their high power density. More important, for side-by-side installation, SINAMICS S200 servo converters require only 1 mm for 200 V series and 10 mm for 400 V series.

Easy to use

Easy setup

All frame sizes of SINAMICS S200 servo converters have an integrated holding brake control [1], which does not require connection to an external relay, and the integrated brake resistor to dissipate the regenerative power for fast braking, which is easy and cost effective for applications.

Easy engineering.

With directly integrated into SINAMICS Startdrive, and the embedded SINAMICS web server for SINAMICS S200, variety of user-friendly features make the servo drive system easy for engineering. Additionally, advanced One Button Tuning (OBT) enables machines to achieve a high dynamic performance within just seconds.

Easy implementation

The system is designed to meet global standards and certifications, including CE, UKCA, UL, RCM, EAC, KC and SEMI F47, ensuring compliance with safety and quality regulations worldwide.

Fit for future

Digital engineering journey

Unlock potential with a seamless digital engineering journey, from selection and commissioning to diagnosis via TIA Selection Tool, Siemens Product Configurator and TIA Portal. More important, SINAMICS S200 servo drive system can be simulated with DriveSim Designer.

Scalable system

As a scalable system, users can get a seamless integrated motion control system together with SIMATIC controllers (e.g. SIMATIC S7-1500).

Safety & Security

The system includes a safety concept that features Safety Integrated Basic Function Safe Torque Off (STO), ensuring safe and reliable operation. You achieve defense in depth thanks to User Management and Access Control (UMAC) providing always active industrial security service.

Siemens EcoTech Profile (SEP) for SINAMICS S200

Siemens introduces a new label: Siemens EcoTech. It is an environmental declaration for products based on product-specific evaluations of sustainability relevant KPIs.

To provide maximum transparency, the Siemens EcoTech Profile (SEP) were created for all Siemens EcoTech products.

It is a Siemens-created, standardized product data sheet which gives insight about the product performance in environmental relevant criteria as well as about interpretation and comparison of data against an existing norm, standard or predecessor product.

You can find more information on the internet at:
https://support.industry.siemens.com/cs/ww/en/ps/29596/cert?ci=5702

www.siemens.com/SiemensEcoTech

Extended warranty

For SINAMICS S200 servo drive system, Siemens offers an optional extended warranty of up to 5½ years via **Service Protect**:

- Free for the first 6 months after registering the product at: www.siemens.com/serviceprotect
- Subject to a charge for a further 3 or 5 years of extended warranty for SINAMICS S200 servo converters.
- Subject to a charge for a further 1 or 2 years of extended warranty for SIMOTICS S-1FL2 servo motors.

You can find detailed information here:
https://support.industry.siemens.com/cs/ww/en/sc/4842

You can find the correct Service Protect product number for your SINAMICS S200 servo drive system using our simple configurator
www.siemens.com/serviceprotect

Concerning standard warranty please ask your partner at Siemens.

Your partner can be found in our Personal Contacts Database at:
www.siemens.com/automation-contact

[1] Not available for SINAMICS S200 Basic servo converters.

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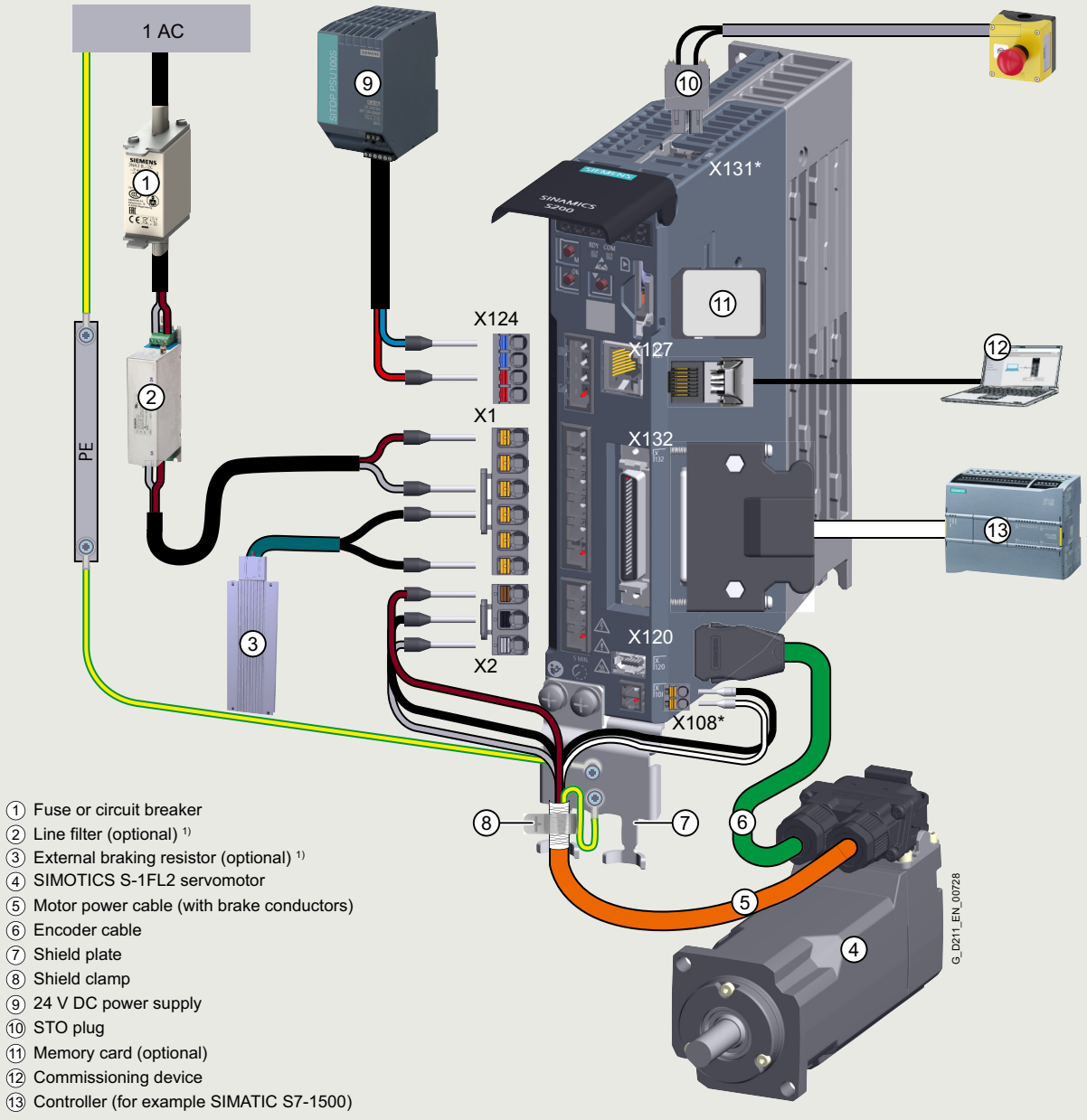
Application

Application examples

	SINAMICS S200 servo drive system Low Inertia for high dynamic performance 200 V ... 240 V 1 AC / 3 AC	Medium Inertia for dynamic and smooth operational performance 200 V ... 240 V 1 AC / 3 AC or 380 V ... 480 V 3 AC	High Inertia for smooth operational performance 380 V ... 480 V 3 AC
Electronics	• LED sorting machine • Glass cover inspection machine • PCB-AOI detection machine • PCB laser cutting machine • Screwing machine • PCB feeding and blanking machine • Assembly production line	• Assembly production line	–
Battery	• Stacking machine • EL filling machine • Busbar welding machine • Case insertion machine • Detection machine	• Laser welding machine • Calendering and slitting machine • Die-cutting machine	• Coating machine • Calendering and slitting machine • Die-cutting machine
Solar	• Stringer machine • Plating diamond machine • Screen printing machine • Auto sorting machine	• Single crystal growing furnace • Batch texturing equipment • Squaring machine	–
Packaging	• Filling and sealing machine • Bag packing machine	• Pillow type packaging machine • Roller type blister packaging machine • Rigid box maker • Labelling machine	• Auto roll-fed square bottom paper bag machine • Rigid box maker
Handling and robotics	• 3- or 4-axis delta picker • 4-axis cylindrical robot • 5-axis roller picker • 4-axis tripod robot	• Cartesian portal	• Truss manipulator
Wood working	• Edging machine • Panel sawing machine	• Wooden engraving • Wooden machine center	• Wooden engraving • Wooden machine center
Metal forming	–	–	• Punching machine • Engraving machine • Edging press machine
Other general machinery	–	• Diaper production line • Stator winding machine • Winding embedding machine • Battery swap station	• Diaper production line • Stator winding machine • Copper foil green foil machine • Battery swap station

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Design



① Fuse or circuit breaker
② Line filter (optional) [1]
③ External braking resistor (optional) [1]
④ SIMOTICS S-1FL2 servomotor
⑤ Motor power cable (with brake conductors)
⑥ Encoder cable
⑦ Shield plate
⑧ Shield clamp
⑨ 24 V DC power supply
⑩ STO plug
⑪ Memory card (optional)
⑫ Commissioning device
⑬ Controller (for example SIMATIC S7-1500)

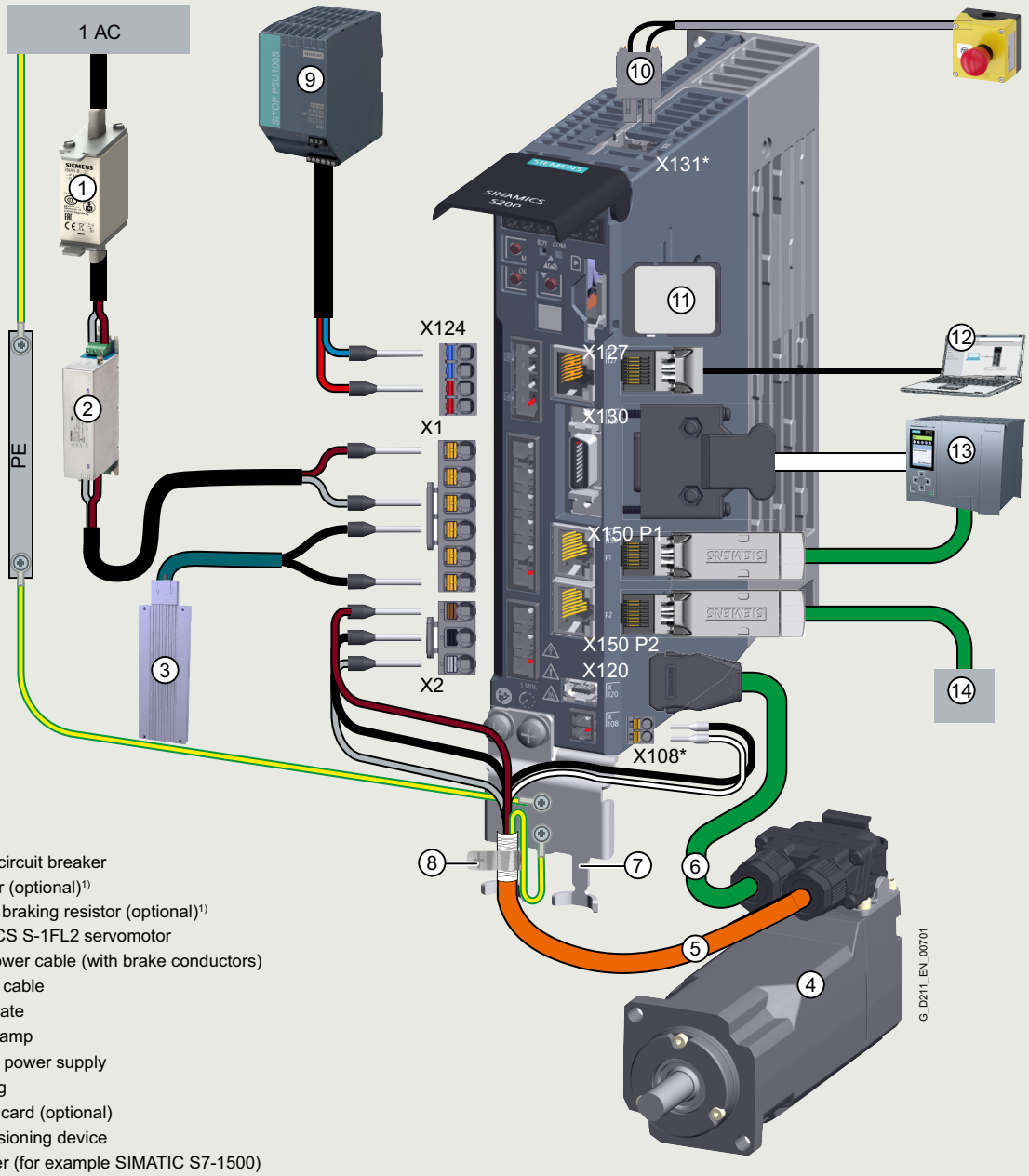
* The interfaces X131 and X108 are only available on the SINAMICS S200 converters, not on the SINAMICS S200 Basic converters.

[1] When using the components, make sure that they are connected to the ground correctly.

System connection overview SINAMICS S200 servo drive system, Pulse Train Input version (PTI), 1 AC

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Design



① Fuse or circuit breaker
② Line filter (optional)[1)]
③ External braking resistor (optional)[1)]
④ SIMOTICS S-1FL2 servomotor
⑤ Motor power cable (with brake conductors)
⑥ Encoder cable
⑦ Shield plate
⑧ Shield clamp
⑨ 24 V DC power supply
⑩ STO plug
⑪ Memory card (optional)
⑫ Commissioning device
⑬ Controller (for example SIMATIC S7-1500)
⑭ PROFINET connection for the next participant

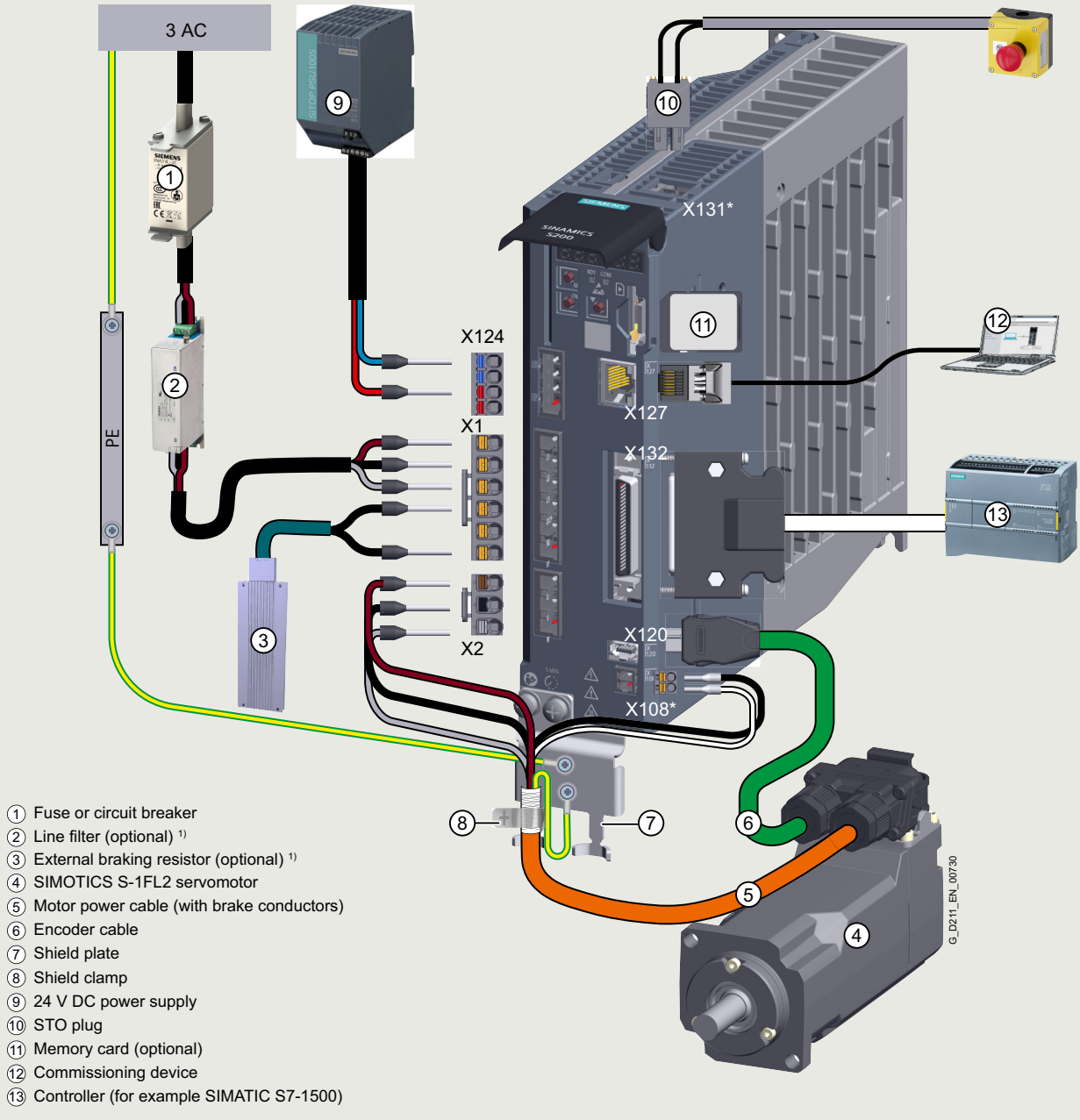
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[1)] When using the components, make sure that they are connected to the ground correctly.

System connection overview SINAMICS S200 servo drive system, PROFINET version, 1 AC

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Design



① Fuse or circuit breaker
② Line filter (optional) 1)
③ External braking resistor (optional) 1)
④ SIMOTICS S-1FL2 servomotor
⑤ Motor power cable (with brake conductors)
⑥ Encoder cable
⑦ Shield plate
⑧ Shield clamp
⑨ 24 V DC power supply
⑩ STO plug
⑪ Memory card (optional)
⑫ Commissioning device
⑬ Controller (for example SIMATIC S7-1500)

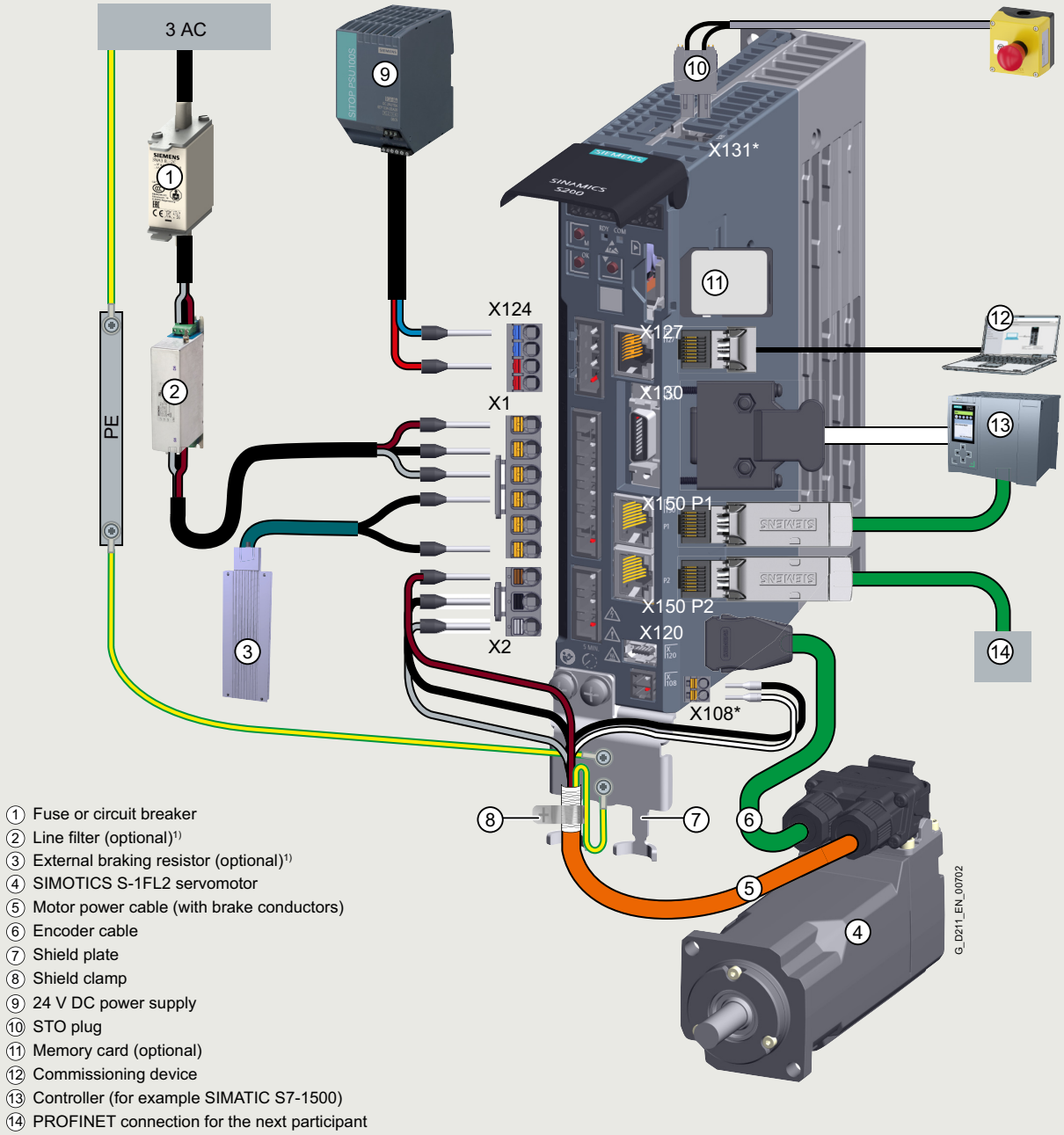
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1) When using the components, make sure that they are connected to the ground correctly.

System connection overview SINAMICS S200 servo drive system, Pulse Train Input version (PTI), 3 AC

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Design



① Fuse or circuit breaker
② Line filter (optional)[1)]
③ External braking resistor (optional)[1)]
④ SIMOTICS S-1FL2 servomotor
⑤ Motor power cable (with brake conductors)
⑥ Encoder cable
⑦ Shield plate
⑧ Shield clamp
⑨ 24 V DC power supply
⑩ STO plug
⑪ Memory card (optional)
⑫ Commissioning device
⑬ Controller (for example SIMATIC S7-1500)
⑭ PROFINET connection for the next participant

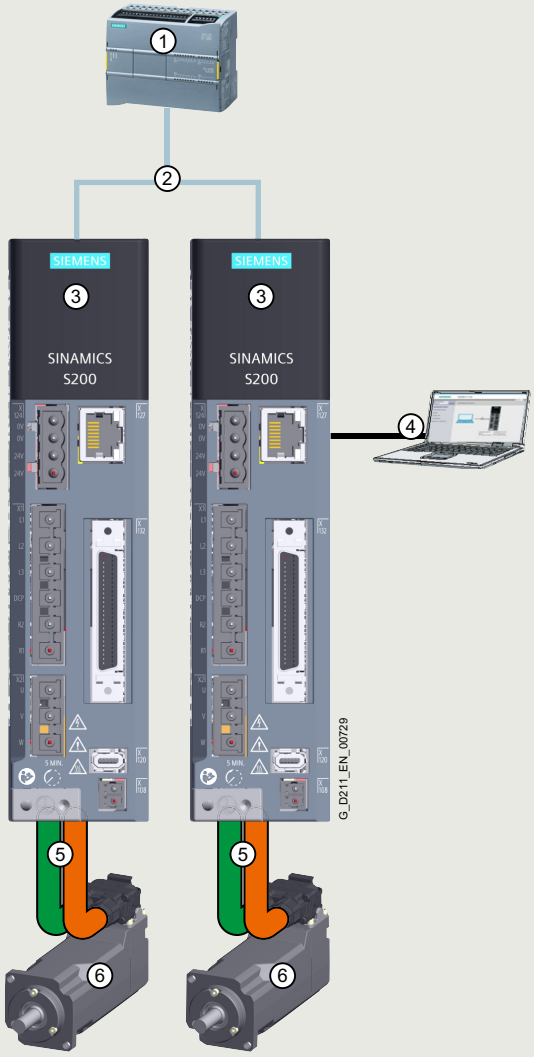
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[1)] When using the components, make sure that they are connected to the ground correctly.

System connection overview SINAMICS S200 servo drive system, PROFINET version, 3 AC

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Integration



① Controller (for example SIMATIC S7-1500)

② Communication between the converter and the controller via setpoint cable

③ SINAMICS S200 converter product line (PTI version)

④ Engineering via LAN with the web server in the converter

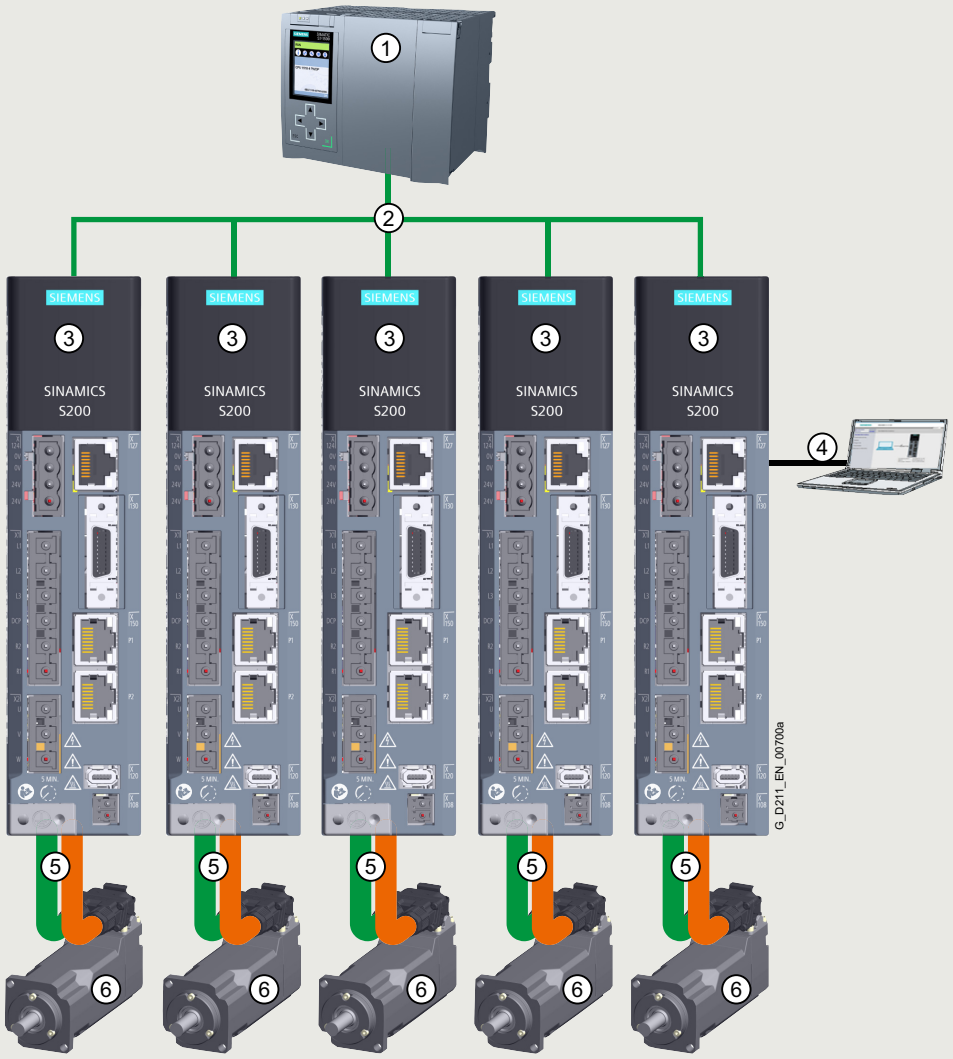
⑤ MOTION-CONNECT 350 or MOTION-CONNECT 380 cables for connecting motor, motor holding brake and encoder

⑥ SIMOTICS S-1FL2 servomotor

System topology SINAMICS S200 servo drive system, Pulse Train Input version (PTI)

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Integration



① Controller (for example SIMATIC S7-1500)

② Communication between the converter and the controller via PROFINET

③ SINAMICS S200 converter product line (PROFINET version)

④ Engineering via LAN with the web server in the converter

⑤ MOTION-CONNECT 350 or MOTION-CONNECT 380 cables for connecting motor, motor holding brake and encoder

⑥ SIMOTICS S-1FL2 servomotor

System topology SINAMICS S200 servo drive system, PROFINET version

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Configuration

The following electronic configuring aids and engineering tools are available for the SINAMICS S200 servo drive system:

DriveSim Designer (firmware V6.2 or higher)

DriveSim Designer provides easy-to-use models for PROFIdrive-enabled SINAMICS converters, so you can create a digital twin of your drive.

More information is provided on the internet at:
www.siemens.com/drive-virtualization

You can find more information on DriveSim Designer in the Engineering tools section.

SINAMICS Selector app

Mobile selection guide for frequency converters

Siemens has developed the SINAMICS Selector app as a practical tool for finding article numbers for your SINAMICS converters in a power range from 0.1 kW to 630 kW quickly and easily. Whether for SINAMICS V20, SINAMICS V90, SINAMICS G120C, SINAMICS G120P, SINAMICS G120X, SINAMICS G120, SINAMICS G220, SINAMICS S200 or SINAMICS S210: the app will provide you with the correct article numbers conveniently.

How does it work?
Simply select your application, the frequency converter you require, the rated power and device options as well as the necessary accessories.

Then you can save your selection and send it by email. Your preselection is the basis for an order specification with the dealer/Siemens.

You will find free downloads for Android and iOS here:
www.siemens.com/sinamics-selector

Siemens Product Configurator

The Siemens Product Configurator helps you configure the optimum drive technology products for a number of applications – starting with gearboxes, motors, converters as well as the associated options and components and ending with controllers, software licenses and connection systems.

The Siemens Product Configurator can be used on the internet without requiring any installation. The
Siemens Product Configurator can be found in SiePortal at the following address:
www.siemens.com/spc

You can find more information on the
Siemens Product Configurator in the Engineering tools section.

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Configuration

Drive dimensioning of the SINAMICS S200 servo drive system with the TIA Selection Tool

The SINAMICS S200 servo drive system is easily configured with the TIA Selection Tool under the Drive Dimensioning plug-in. It provides support when selecting the hardware and firmware components necessary to implement a drive task. The plug-in encompasses the configuration of the entire drive system and allows the handling of individual drives.

- Intuitive user interface, menu-based operation and help
- Configuration of the SINAMICS S200 servo drive system
- Adjustable traversing cycles and various mechanical systems integrated for single-axis applications
- Interface to the TIA Portal and SiePortal

The TIA Selection Tool is available for free on the internet at www.siemens.com/tia-selection-tool-standalone

You can find more information on the TIA Selection Tool in the Engineering tools section.

SINAMICS web server for SINAMICS S200

The converter can be brought easily and quickly into operation and optimized with the web server. Additional software does not need to be installed as access comes directly from a web browser.

You can find more information about the SINAMICS web server for SINAMICS S200 in the Engineering tools section.

SINAMICS Startdrive commissioning tool (TIA Portal)

SINAMICS Startdrive is a tool for configuring, commissioning, and diagnosing the SINAMICS converter family and is integrated into TIA Portal (V18 SP2 or higher).

The SINAMICS Startdrive has been optimized with regard to user friendliness and consistent use of the TIA Portal benefits of a common working environment for PLC, HMI and drives. SINAMICS firmware V6.2 or higher is required for SINAMICS S200.

The SINAMICS Startdrive Advanced license is required for the Safety Integrated acceptance test.

The SINAMICS Startdrive commissioning tool is available for free on the internet at:
www.siemens.com/startdrive

You can find more information on the SINAMICS Startdrive commissioning tool in the Engineering tools section.

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Technical specifications

General technical specifications

	SINAMICS S200 servo drive system Low Inertia for high dynamic performance 200 V ... 240 V 1 AC / 3 AC	Medium Inertia for dynamic and smooth operational performance 200 V ... 240 V 1 AC / 3 AC or 380 V ... 480 V 3 AC	High Inertia for smooth operational performance 380 V ... 480 V 3 AC
SINAMICS S200 Basic servo converters			
Line voltage and power range	• 200 V … 240 V 1 AC / 3 AC (–15 % / +10 %), 0.1 kW … 1 kW	• 200 V … 240 V 1 AC / 3 AC (–15 % / +10 %), 0.1 kW … 1 kW	–
Communication and control type	• PROFINET RT/IRT (PN) for PROFINET version (PN): Speed control (TO), basic positioner control (EPOS) • Pulse Train Input version (PTI): Fast position control with pulse train input (Fast PTI)		–
Inputs and outputs	• PROFINET version (PN): 4 DI (incl. 2 fast DI), 1 DO • Pulse Train Input version (PTI): 6 DI, 4 DO		–
Frame sizes	• FSA, FSB, FSC		–
Safety Integrated Basic Functions	–		–
Holding brake control	• External relay and power supply required		–
Degree of protection	• IP20		–
SINAMICS S200 servo converters			
Line voltage and power range	• 200 V … 240 V 1 AC / 3 AC (–15 % / +10 %), 0.1 kW … 1 kW	• 200 V … 240 V 1 AC / 3 AC (–15 % / +10 %), 0.1 kW … 1 kW • 380 V … 480 V 3 AC (–15 % / +10 %), 0.2 kW … 1.5 kW	• 380 V … 480 V 3 AC (–15 % / +10 %), 0.4 kW … 7 kW
Communication and control type	• PROFINET RT/IRT (PN) for PROFINET version (PN): Speed control (TO), basic positioner control (EPOS) • Modbus RTU for Pulse Train Input version (PTI): Fast position control with pulse train input (Fast PTI), speed control (S), torque control (T), compound speed/torque control (S/T)		
Inputs and outputs	• PROFINET version (PN): 4 DI (incl. 2 fast DI), 2 DO • Pulse Train Input version (PTI): 10 DI (incl. 2 fast DI), 6 DO, 2 AI, 2 AO		
Frame sizes	• FSA, FSB, FSC		• FSA, FSB, FSC, FSD
Safety Integrated Basic Functions	• Safe Torque Off (STO)		
Degree of protection	• IP20		
Holding brake control	• Integrated holding brake control		
SIMOTICS S-1FL2 servomotors			
Shaft height	• 20, 30, 40, 40 Compact, 52	• 30, 40, 40 Compact, 48	• 45, 65, 90
Rated torque	• 0.16 ... 6.60 Nm (0.12 … 4.87 lbf-ft)	• 0.64 ... 4.60 Nm (0.47 … 3.39 lbf-ft)	• 1.27 ... 33.40 Nm (0.94 … 24.63 lbf-ft)
Rated power	• 0.05 ... 2.1 kW	• 0.2 ... 1.45 kW	• 0.4 ... 7 kW
Rated speed	• 3000 r/min		• 3000 r/min for shaft height 45 • 2000 r/min for shaft heights 65, 90
Max. speed	• 6500 r/min for shaft heights 20, 30, 40, 40 Compact • 6000 r/min for shaft height 52	• 6500 r/min for shaft heights 30, 40, 40 Compact • 6000 r/min for shaft height 48	• 4000 r/min for shaft height 45 • 3000 r/min for shaft height 65 • 2000 r/min, 2500 r/min or 3000 r/min for shaft height 90
Encoder	• Absolute encoder 17-bit singleturn • Absolute encoder 21-bit singleturn • Absolute encoder 21-bit singleturn + 12-bit multiturn		
Degree of protection	• IP54 for shaft heights 20, 30, 40 Optional: IP65 • IP65, natural cooling, with shaft sealing ring for shaft heights 40 Compact, 52	• IP54 for shaft heights 30, 40 Optional: IP65 • IP65, natural cooling, with shaft sealing ring for shaft heights 40 Compact, 48	• IP65, natural cooling, with shaft sealing ring
Connection technology	• Metal hybrid connector for shaft heights 20, 30, 40, 40 Compact • Metal circular connector for shaft height 52	• Metal hybrid connector for shaft heights 30, 40, 40 Compact • Metal circular connector for shaft height 48	• Metal circular connector
MOTION-CONNECT 350 and 380 connection systems			
Cable	• Pre-assembled dual cables for power and signals for shaft heights 20, 30, 40, 40 Compact • Pre-assembled power and signal cables for shaft height 52	• Pre-assembled dual cables for power and signals for shaft heights 30, 40, 40 Compact • Pre-assembled power and signal cables for shaft height 48	• Pre-assembled power cables • Pre-assembled encoder cables • Pre-assembled brake cables
Length	3 m (9.84 ft), 5 m (16.4 ft), 7 m (23.0 ft), 10 m (32.8 ft), 15 m (49.2 ft), 20 m (65.6 ft), 30 m (98,4 ft)		
Outlet direction	Shaft heights 20, 30, 40, 40 Compact: • Drive end (DE) • Non drive end (NDE) Shaft height 52: • Rotatable	Shaft heights 30, 40, 40 Compact: • Drive end (DE) • Non drive end (NDE) Shaft height 48: • Rotatable	• Rotatable

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Clicking to SiePortal

6SL3255-0AA00-5AA0



SINAMICS S200 servo drive system with SIMOTICS S-1FL2 Low Inertia servomotors

Selection and ordering data

SIMOTICS S-1FL2 servomotors – Further info in section "SIMOTICS S-1FL2 servomotors".

Max. speed	Rated power [1]	Static torque	Rated torque [1]	Max. torque [1]	Rated current	Max. current		Torque constant	Moment of inertia of rotor		Recommended load to motor inertia ratio, max.	Weight [2]	
$n_{max.}$	P_N at ΔT=100 K	M_0 at ΔT=100 K	M_N at ΔT=100 K	M_{max}	I_N at ΔT=100 K	I_{max}			$J_{without\ brake}$	$J_{with\ brake}$		$m_{without\ brake}$	$m_{with\ brake}$
r/min	kW (hp)	Nm	Nm	Nm	A	A	Article No.	Nm/A	kg cm^2	kg cm^2		kg	kg
SIMOTICS S-1FL2 Low Inertia servomotors – For high dynamic performance													
Shaft height 20 – Rated speed n_N 3000 r/min – Line voltage 200 ... 240 V 1 AC / 3 AC													
6500	0.05 (0.07)	0.16	0.16	0.56	0.81	3.1	**1FL2102-2AG ■ ■ - ■ ■ C 0**	0.2	0.017	0.022	40×	0.29	0.46
6500	0.10 (0.14)	0.32	0.32	1.12	0.76	2.95	**1FL2102-4AG ■ ■ - ■ ■ C 0**	0.42	0.029	0.034	30×	0.39	0.56
Shaft height 20 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6500	0.05 (0.07)	0.16	0.16	0.56	0.84	3.4	**1FL2102-2AF ■ ■ - ■ ■ C 0**	0.19	0.017	0.022	40×	0.29	0.45
6500	0.10 (0.14)	0.32	0.32	1.12	0.81	2.9	**1FL2102-4AF ■ ■ - ■ ■ C 0**	0.4	0.029	0.034	30×	0.38	0.55
Shaft height 30 – Rated speed n_N 3000 r/min – Line voltage 200 ... 240 V 1 AC / 3 AC													
6500	0.20 (0.27)	0.64	0.64	2.24	1.28	4.8	**1FL2103-2AG ■ ■ - ■ ■ C 0**	0.5	0.08	0.11	30×	0.79	1.14
6500	0.40 (0.54)	1.27	1.27	4.45	2.4	8.7	**1FL2103-4AG ■ ■ - ■ ■ C 0**	0.53	0.155	0.185		1.16	1.5
Shaft height 30 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6500	0.20 (0.27)	0.64	0.64	2.24	1.3	4.8	**1FL2103-2AF ■ ■ - ■ ■ C 0**	0.49	0.08	0.11	30×	0.78	1.12
6500	0.40 (0.54)	1.27	1.27	4.45	1.3	4.8	**1FL2103-4AF ■ ■ - ■ ■ C 0**	0.98	0.155	0.185		1.14	1.49
Shaft height 40 or 40 Compact – Rated speed n_N 3000 r/min – Line voltage 200 ... 240 V 1 AC / 3 AC													
6500	0.75 (1.02)	2.4	2.4	8	4.4	16	**1FL2104-2AG ■ ■ - ■ ■ C 0**	0.55	0.5	0.58	20×	2.7	3.5
6500	0.75 (1.02)	2.4	2.4	8.4	4.2	16	**1FL2104-2BG ■ 1 - ■ ■ C 0**	0.57	0.67	0.77		2.0	2.8
6500	1.00 (1.36)	3.2	3.2	10.8	4.4	16	**1FL2104-4AG ■ ■ - ■ ■ C 0**	0.73	0.63	0.71		3.3	4.1
6500	1.00 (1.36)	3.2	3.2	11.2	4.4	16	**1FL2104-4BG ■ 1 - ■ ■ C 0**	0.73	0.87	0.97		2.4	3.2
Shaft height 40 or 40 Compact – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6500	0.75 (1.02)	2.4	2.4	8	2.1	7.6	**1FL2104-2AF ■ ■ - ■ ■ C 0**	1.14	0.5	0.58	20×	2.7	3.5
6500	0.75 (1.02)	2.4	2.4	8.4	2.4	9.3	**1FL2104-2BF ■ 1 - ■ ■ C 0**	1.0	0.67	0.77		2.0	2.8
6500	1.00 (1.36)	3.2	3.2	10.8	3	10.9	**1FL2104-4AF ■ ■ - ■ ■ C 0**	1.07	0.63	0.71		3.3	4.1
6500	1.00 (1.36)	3.2	3.2	11.2	3	10.9	**1FL2104-4BF ■ 1 - ■ ■ C 0**	1.07	0,87	0,97		2.4	3.2
Shaft height 52 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6000	1.45 (2.04)	5	4.6	15.9	4.35	18	**1FL2105-2AF ■ 1 - ■ ■ B 0**	1.08	1.71	2.54	20×	5.5	6.4
6000	2.10 (2.72)	8	6.6	25.3	5.6	24	**1FL2105-4AF ■ 1 - ■ ■ B 0**	1.19	2.65	3.48	15×	7.8	8.7

Brake					
Without	**0**				
Holding brake	**1**				
Degree of protection					
IP54		**0**			
IP65 (with shaft sealing ring)		**1**			
Shaft end, feather key					
Plain shaft			**0**		
Shaft with feather key (half-key balancing)			**1**		
Encoder					
Absolute encoder 17-bit singleturn				**H**	
Absolute encoder 21-bit singleturn				**S**	
Absolute encoder 21-bit singleturn + 12-bit multiturn				**M**	
Connection technology					
Metal circular connector					**B**
Metal hybrid connector					**C**

In addition, the Siemens Product Configurator can be used on the internet: www.siemens.com/sinamics-s200/configuration.

For footnotes, see next page.

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SINAMICS S200 servo converters Further info in section "SINAMICS S200 servo converters".			Line filter			Recommended standard fuse More information at www.siemens.com/sinamics-s200/ocpd				Recommended circuit breaker
Max. motor power		Frame size	With one of the recommended line filters, IEC 61800-3 category C2 can be reached in combination with SINAMICS S200 for motor cable lengths up to 30 m (98.5 ft).			IEC-compliant		UL/cUL-compliant		IEC/UL/cUL-compliant
			Line voltage	I_N		I	3NA3	I	Fuse type	3RV2
kW (hp)	Article No.		V	A	Article No.	A	Article No.	A	Class	Article No.
SINAMICS S200 and SINAMICS S200 Basic servo converters										
Line voltage 200 ... 240 V 1 AC / 3 AC										
0.10 (0.14)	**6SL5 ■ 10-1BB10-1A ■ 0**	FSA	1 AC 200 ... 240	18	**6SL5550-0XB21-8CA0**	4	**3NA3804**	3	J	**3RV2011-1EA10**
			3 AC 200 ... 240	5	**6SL3203-0BE15-0VA0**					
Line voltage 380 ... 480 V 3 AC										
0.20 (0.27)	**6SL5 5 10-1BE10-2A ■ 0**	FSA	3 AC 380 ... 480	5	**6SL5550-0XE20-5CA0**	4	**3NA3804**	3	J	**3RV2011-1CA10**
Line voltage 200 ... 240 V 1 AC / 3 AC										
0.20 (0.27)	**6SL5 ■ 10-1BB10-2A ■ 0**	FSA	1 AC 200 ... 240	18	**6SL5550-0XB21-8CA0**	4	**3NA3804**	3	J	**3RV2011-1EA10**
			3 AC 200 ... 240	5	**6SL3203-0BE15-0VA0**					
0.40 (0.54)	**6SL5 ■ 10-1BB10-4A ■ 0**	FSB	1 AC 200 ... 240	18	**6SL5550-0XB21-8CA0**	6	**3NA3801**	6	J	**3RV2011-1HA10**
			3 AC 200 ... 240	5	**6SL3203-0BE15-0VA0**					**3RV2011-1EA10**
Line voltage 380 ... 480 V 3 AC										
0.20 (0.27)	**6SL5 5 10-1BE10-2A ■ 0**	FSA	3 AC 380 ... 480	5	**6SL5550-0XE20-5CA0**	4	**3NA3804**	3	J	**3RV2011-1CA10**
0.40 (0.54)	**6SL5 5 10-1BE10-4A ■ 0**									**3RV2021-1DA10**
Line voltage 200 ... 240 V 1 AC / 3 AC										
0.75 (1.02)	**6SL5 ■ 10-1BB10-8A ■ 0**	FSC	1 AC 200 ... 240	18	**6SL5550-0XB21-8CA0**	16	**3NA3805**	15	J	**3RV2011-1KA10**
			3 AC 200 ... 240	5	**6SL3203-0BE15-0VA0**					**3RV2011-1HA10**
1.00 (1.36)	**6SL5 ■ 10-1BB11-0A ■ 0**	FSC	1 AC 200 ... 240	18	**6SL5550-0XB21-8CA0**	16	**3NA3805**	15	J	**3RV2011-1KA10**
			3 AC 200 ... 240	5	**6SL3203-0BE15-0VA0**					**3RV2011-1JA10**
Line voltage 380 ... 480 V 3 AC										
0.75 (1.02)	**6SL5 5 10-1BE10-8A ■ 0**	FSB	3 AC 380 ... 480	5	**6SL5550-0XE20-5CA0**	10	**3NA3803**	10	J	**3RV2021-1EA10**
1.00 (1.36)	**6SL5 5 10-1BE11-0A ■ 0**									**3RV2021-1FA10**
Line voltage 380 ... 480 V 3 AC										
1.75 (2.38)	**6SL5 5 10-1BE11-5A ■ 0**	FSC	3 AC 380 ... 480	12	**6SL5550-0XE21-2CA0**	16	**3NA3805**	15	J	**3RV2011-4AA10**
2.50 (3.40)	**6SL5 5 10-1BE12-5A ■ 0**							20		**3RV2011-4AA10**

Servo converter	
SINAMICS S200	**5**
SINAMICS S200 Basic	**6**
Version	
Pulse Train Input (PTI)	**A**
PROFINET (PN)	**F**

[1] Rated torque, rated power and maximum torque listed in the table above allow for a production tolerance of 10 %.

[2] Motor weight with absolute encoder 17-bit singleturn.

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Selection and ordering data

SIMOTICS S-1FL2 servomotors – Further info in section "SIMOTICS S-1FL2 servomotors"

Max. speed	Rated power [1]	Static torque	Rated torque [1]	Max. torque [1]	Rated current	Max. current		Torque constant	Moment of inertia of rotor		Recommended load to motor inertia ratio, max.	Weight [2]	
$n_{max.}$	P_N at ΔT=100 K	M_0 at ΔT=100 K	M_N at ΔT=100 K	M_{max}	I_N at ΔT=100 K	I_{max}			$J_{without\ brake}$	$J_{with\ brake}$		$m_{without\ brake}$	$m_{with\ brake}$
r/min	kW (hp)	Nm	Nm	Nm	A	A	Article No.	Nm/A	kg cm²	kg cm²		kg	kg
SIMOTICS S-1FL2 Low Inertia servomotors – For high dynamic performance													
Shaft height 20 – Rated speed n_N 3000 r/min – Line voltage 200 ... 240 V 1 AC / 3 AC													
6500	0.05 (0.07)	0.16	0.16	0.56	0.81	3.1	**1FL2102-2AG ■ ■ - ■ ■ C 0**	0.2	0.017	0.022	40×	0.29	0.46
6500	0.10 (0.14)	0.32	0.32	1.12	0.76	2.95	**1FL2102-4AG ■ ■ - ■ ■ C 0**	0.42	0.029	0.034	30×	0.39	0.56
Shaft height 20 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6500	0.05 (0.07)	0.16	0.16	0.56	0.84	3.4	**1FL2102-2AF ■ ■ - ■ ■ C 0**	0.19	0.017	0.022	40×	0.29	0.45
6500	0.10 (0.14)	0.32	0.32	1.12	0.81	2.9	**1FL2102-4AF ■ ■ - ■ ■ C 0**	0.4	0.029	0.034	30×	0.38	0.55
Shaft height 30 – Rated speed n_N 3000 r/min – Line voltage 200 ... 240 V 1 AC / 3 AC													
6500	0.20 (0.27)	0.64	0.64	2.24	1.28	4.8	**1FL2103-2AG ■ ■ - ■ ■ C 0**	0.5	0.08	0.11	30×	0.79	1.14
6500	0.40 (0.54)	1.27	1.27	4.45	2.4	8.7	**1FL2103-4AG ■ ■ - ■ ■ C 0**	0.53	0.155	0.185		1.16	1.5
Shaft height 30 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6500	0.20 (0.27)	0.64	0.64	2.24	1.3	4.8	**1FL2103-2AF ■ ■ - ■ ■ C 0**	0.49	0.08	0.11	30×	0.78	1.12
6500	0.40 (0.54)	1.27	1.27	4.45	1.3	4.8	**1FL2103-4AF ■ ■ - ■ ■ C 0**	0.98	0.155	0.185		1.14	1.49
Shaft height 40 or 40 Compact – Rated speed n_N 3000 r/min – Line voltage 200 ... 240 V 1 AC / 3 AC													
6500	0.75 (1.02)	2.4	2.4	8	4.4	16	**1FL2104-2AG ■ ■ - ■ ■ C 0**	0.55	0.5	0.58	20×	2.7	3.5
6500	0.75 (1.02)	2.4	2.4	8.4	4.2	16	**1FL2104-2BG ■ 1 - ■ ■ C 0**	0.57	0.67	0.77		2.0	2.8
6500	1.00 (1.36)	3.2	3.2	10.8	4.4	16	**1FL2104-4AG ■ ■ - ■ ■ C 0**	0.73	0.63	0.71		3.3	4.1
6500	1.00 (1.36)	3.2	3.2	11.2	4.4	16	**1FL2104-4BG ■ 1 - ■ ■ C 0**	0.73	0.87	0.97		2.4	3.2
Shaft height 40 or 40 Compact – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6500	0.75 (1.02)	2.4	2.4	8	2.1	7.6	**1FL2104-2AF ■ ■ - ■ ■ C 0**	1.14	0.5	0.58	20×	2.7	3.5
6500	0.75 (1.02)	2.4	2.4	8.4	2.4	9.3	**1FL2104-2BF ■ 1 - ■ ■ C 0**	1.0	0.67	0.77		2.0	2.8
6500	1.00 (1.36)	3.2	3.2	10.8	3	10.9	**1FL2104-4AF ■ ■ - ■ ■ C 0**	1.07	0.63	0.71		3.3	4.1
6500	1.00 (1.36)	3.2	3.2	11.2	3	10.9	**1FL2104-4BF ■ 1 - ■ ■ C 0**	1.07	0,87	0,97		2.4	3.2
Shaft height 52 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6000	1.45 (2.04)	5	4.6	15.9	4.35	18	**1FL2105-2AF ■ 1 - ■ ■ B 0**	1.08	1.71	2.54	20×	5.5	6.4
6000	2.10 (2.72)	8	6.6	25.3	5.6	24	**1FL2105-4AF ■ 1 - ■ ■ B 0**	1.19	2.65	3.48	15×	7.8	8.7

Brake						
Without	**0**					
Holding brake	**1**					
Degree of protection						
IP54		**0**				
IP65 (with shaft sealing ring)		**1**				
Shaft end, feather key						
Plain shaft			**0**			
Shaft with feather key (half-key balancing)			**1**			
Encoder						
Absolute encoder 17-bit singleturn				**H**		
Absolute encoder 21-bit singleturn				**S**		
Absolute encoder 21-bit singleturn + 12-bit multiturn				**M**		
Connection technology						
Metal circular connector					**B**	
Metal hybrid connector					**C**	

In addition, the Siemens Product Configurator can be used on the internet: www.siemens.com/sinamics-s200/configuration

[1] Rated torque, rated power and maximum torque listed in the table above allow for a production tolerance of 10 %.

[2] Motor weight with absolute encoder 17-bit singleturn.

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	MOTION-CONNECT connection systems – Further info in section "MOTION-CONNECT 350 and 380 connection systems".					
	Pre-assembled dual cables for power and communication		Pre-assembled power cables		Pre-assembled cables for communication	
					SINAMICS S200 – Absolute encoder on the 1FL2 servomotor	
	No. of cores × cross-section mm²	Article No.	No. of cores × cross-section mm²	Article No.	No. of cores × cross-section mm²	Article No.
	MOTION-CONNECT connection systems					
	4 × 0.75 / 3 × 2 × 0.25 [1] or 4 × 0.75 / 3 × 2 × 0.25 [1] / 2 × 0.38	**6FX3 ■02-7 ■ ■01-1 ■ ■0**	–	–	–	–
	4 × 0.75 / 3 × 2 × 0.25 [1] or 4 × 0.75 / 3 × 2 × 0.25 [1] / 2 × 0.38	**6FX3 ■02-7 ■ ■01-1 ■ ■0**	–	–	–	–
	4 × 0.75 / 3 × 2 × 0.25 [1] or 4 × 0.75 / 3 × 2 × 0.25 [1] / 2 × 0.38	**6FX3 ■02-7 ■ ■01-1 ■ ■0**	–	–	–	–
	4 × 0.75 / 3 × 2 × 0.25 [1] or 4 × 0.75 / 3 × 2 × 0.25 [1] / 2 × 0.38	**6FX3 ■02-7 ■ ■01-1 ■ ■0**	–	–	–	–
	4 × 0.75 / 3 × 2 × 0.25 [1] or 4 × 0.75 / 3 × 2 × 0.25 [1] / 2 × 0.38	**6FX3 ■02-7 ■ ■01-1 ■ ■0**	–	–	–	–
	4 × 0.75 / 3 × 2 × 0.25 [1] or 4 × 0.75 / 3 × 2 × 0.25 [1] / 2 × 0.38	**6FX3 ■02-7 ■ ■01-1 ■ ■0**	–	–	–	–
	–	–	4 × 0.75 or 4 × 0.75 / 2 × 0.38	**6FX3 ■02-5 ■ K32-1 ■ ■0**	3 × 2 × 0.25 [1]	**6FX3 ■02-2XB11-1 ■ ■0**

Connection system																								
MOTION-CONNECT 350	**5**									**5**										**5**				
MOTION-CONNECT 380	**8**									**8**										**8**				
Brake cores																								
Without		**C**										**C**												
With		**D**										**D**												
Outlet direction																								
Drive end (DE)			**D**																					
Non drive end (NDE)			**N**																					
Length																								
3 m				**A**	**D**										**A**	**D**						**A**	**D**	
5 m				**A**	**F**										**A**	**F**						**A**	**F**	
7 m				**B**	**H**										**B**	**H**						**B**	**H**	
10 m				**C**	**A**										**C**	**A**						**C**	**A**	
15 m				**D**	**F**										**D**	**F**						**D**	**F**	
20 m					**A**											**A**							**A**	
30 m					**A**											**A**							**A**	

[1] No. of cores × cross-section for encoder cable in mm² for:
Length <20 m: 3 x 2 x 0.25
Length ≥ 20 m: 2 x 2 x 0.25 + 2 x 0.38

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Selection and ordering data

SIMOTICS S-1FL2 servomotors – Further info in section "SIMOTICS S-1FL2 servomotors"

Max. speed	Rated power [1]	Static torque	Rated torque [1]	Max. torque [1]	Rated current	Max. current		Torque constant	Moment of inertia of rotor		Recommended load to motor inertia ratio, max.	Weight [2]	
$n_{max.}$	P_N at ΔT=100 K	M_0 at ΔT=100 K	M_N at ΔT=100 K	M_{max}	I_N at ΔT=100 K	I_{max}			$J_{without\ brake}$	$J_{with\ brake}$		$m_{without\ brake}$	$m_{with\ brake}$
r/min	kW (hp)	Nm	Nm	Nm	A	A	Article No.	Nm/A	kg cm²	kg cm²		kg	kg
SIMOTICS S-1FL2 Low Inertia servomotors – For high dynamic performance													
Shaft height 20 – Rated speed n_N 3000 r/min – Line voltage 200 ... 240 V 1 AC / 3 AC													
6500	0.05 (0.07)	0.16	0.16	0.56	0.81	3.1	**1FL2102-2AG ■ ■ - ■ ■ C 0**	0.2	0.017	0.022	40×	0.29	0.46
6500	0.10 (0.14)	0.32	0.32	1.12	0.76	2.95	**1FL2102-4AG ■ ■ - ■ ■ C 0**	0.42	0.029	0.034	30×	0.39	0.56
Shaft height 20 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6500	0.05 (0.07)	0.16	0.16	0.56	0.84	3.4	**1FL2102-2AF ■ ■ - ■ ■ C 0**	0.19	0.017	0.022	40×	0.29	0.45
6500	0.10 (0.14)	0.32	0.32	1.12	0.81	2.9	**1FL2102-4AF ■ ■ - ■ ■ C 0**	0.4	0.029	0.034	30×	0.38	0.55
Shaft height 30 – Rated speed n_N 3000 r/min – Line voltage 200 ... 240 V 1 AC / 3 AC													
6500	0.20 (0.27)	0.64	0.64	2.24	1.28	4.8	**1FL2103-2AG ■ ■ - ■ ■ C 0**	0.5	0.08	0.11	30×	0.79	1.14
6500	0.40 (0.54)	1.27	1.27	4.45	2.4	8.7	**1FL2103-4AG ■ ■ - ■ ■ C 0**	0.53	0.155	0.185		1.16	1.5
Shaft height 30 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6500	0.20 (0.27)	0.64	0.64	2.24	1.3	4.8	**1FL2103-2AF ■ ■ - ■ ■ C 0**	0.49	0.08	0.11	30×	0.78	1.12
6500	0.40 (0.54)	1.27	1.27	4.45	1.3	4.8	**1FL2103-4AF ■ ■ - ■ ■ C 0**	0.98	0.155	0.185		1.14	1.49
Shaft height 40 or 40 Compact – Rated speed n_N 3000 r/min – Line voltage 200 ... 240 V 1 AC / 3 AC													
6500	0.75 (1.02)	2.4	2.4	8	4.4	16	**1FL2104-2AG ■ ■ - ■ ■ C 0**	0.55	0.5	0.58	20×	2.7	3.5
6500	0.75 (1.02)	2.4	2.4	8.4	4.2	16	**1FL2104-2BG ■ 1 - ■ ■ C 0**	0.57	0.67	0.77		2.0	2.8
6500	1.00 (1.36)	3.2	3.2	10.8	4.4	16	**1FL2104-4AG ■ ■ - ■ ■ C 0**	0.73	0.63	0.71		3.3	4.1
6500	1.00 (1.36)	3.2	3.2	11.2	4.4	16	**1FL2104-4BG ■ 1 - ■ ■ C 0**	0.73	0.87	0.97		2.4	3.2
Shaft height 40 or 40 Compact – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6500	0.75 (1.02)	2.4	2.4	8	2.1	7.6	**1FL2104-2AF ■ ■ - ■ ■ C 0**	1.14	0.5	0.58	20×	2.7	3.5
6500	0.75 (1.02)	2.4	2.4	8.4	2.4	9.3	**1FL2104-2BF ■ 1 - ■ ■ C 0**	1.0	0.67	0.77		2.0	2.8
6500	1.00 (1.36)	3.2	3.2	10.8	3	10.9	**1FL2104-4AF ■ ■ - ■ ■ C 0**	1.07	0.63	0.71		3.3	4.1
6500	1.00 (1.36)	3.2	3.2	11.2	3	10.9	**1FL2104-4BF ■ 1 - ■ ■ C 0**	1.07	0,87	0,97		2.4	3.2
Shaft height 52 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6000	1.45 (2.04)	5	4.6	15.9	4.35	18	**1FL2105-2AF ■ 1 - ■ ■ B 0**	1.08	1.71	2.54	20×	5.5	6.4
6000	2.10 (2.72)	8	6.6	25.3	5.6	24	**1FL2105-4AF ■ 1 - ■ ■ B 0**	1.19	2.65	3.48	15×	7.8	8.7

Brake						
Without	**0**					
Holding brake	**1**					
Degree of protection						
IP54		**0**				
IP65 (with shaft sealing ring)		**1**				
Shaft end, feather key						
Plain shaft			**0**			
Shaft with feather key (half-key balancing)			**1**			
Encoder						
Absolute encoder 17-bit singleturn				**H**		
Absolute encoder 21-bit singleturn				**S**		
Absolute encoder 21-bit singleturn + 12-bit multiturn				**M**		
Connection technology						
Metal circular connector					**B**	
Metal hybrid connector					**C**	

In addition, the Siemens Product Configurator can be used on the internet: www.siemens.com/sinamics-s200/configuration

[1] Rated torque, rated power and maximum torque listed in the table above allow for a production tolerance of 10 %.

[2] Motor weight with absolute encoder 17-bit singleturn.

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	MOTION-CONNECT connection systems (continued) – Further info in section "MOTION-CONNECT 350 and 380 connection systems".						
	Pre-assembled cables for communication		Connectors				
	SINAMICS S200 – Brake on the 1FL2 servomotor with holding brake		Drive end (DE) direction	Non drive end (NDE) direction	Motor side		
	No. of cores × cross-section mm²	Article No.	for hybrid connection Article No.	for hybrid connection Article No.	for power connection Article No.	for encoder connection Article No.	for brake Article No.
	MOTION-CONNECT connection systems (continued)						
	–	–	**6FX2003-3HD00**	**6FX2003-3HN00**	–	–	–
	–	–	**6FX2003-3HD00**	**6FX2003-3HN00**	–	–	–
	–	–	**6FX2003-3HD00**	**6FX2003-3HN00**	–	–	–
	–	–	**6FX2003-3HD00**	**6FX2003-3HN00**	–	–	–
	–	–	**6FX2003-3HD00**	**6FX2003-3HN00**	–	–	–
	–	–	**6FX2003-3HD00**	**6FX2003-3HN00**	–	–	–
	–	–	–	–	**6FX2003-0LL63**	**6FX2003-0DB63**	– (in power connector)

MOTION-CONNECT connectors for SINAMICS S200 servo converters				
Crimp contacts "power and PE" (2000 pieces per reel) for hybrid connection	Crimp contacts "signal and brake" (2000 pieces per reel) for hybrid connection	Connectors converter side for power connection	IX signal connector, C coded for encoder connection	for brake
6FX2003-7HP00	**6FX2003-7HS00**	–	**6FX2003-0DE61**	–

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SINAMICS S200 servo drive system with SIMOTICS S-1FL2 Medium Inertia servomotors

Selection and ordering data

SIMOTICS S-1FL2 servomotors – Further info in section "SIMOTICS S-1FL2 servomotors".

Max. speed	Rated power [1]	Static torque	Rated torque [1]	Max. torque [1]	Rated current	Max. current		Torque constant	Moment of inertia of rotor		Recommended load to motor inertia ratio, max.	Weight [2]	
$n_{max.}$	P_N at ΔT=100 K	M_0 at ΔT=100 K	M_N at ΔT=100 K	M_{max}	I_N at ΔT=100 K	I_{max}			$J_{without\ brake}$	$J_{with\ brake}$		$m_{without\ brake}$	$m_{with\ brake}$
r/min	kW (hp)	Nm	Nm	Nm	A	A	Article No.	Nm/A	kg cm^2	kg cm^2		kg	kg
SIMOTICS S-1FL2 Medium Inertia servomotors – For dynamic and smooth operational performance													
Shaft height 30 – Rated speed n_N 3000 r/min – Line voltage 200 ... 240 V 1 AC / 3 AC													
6500	0.20 (0.27)	0.64	0.64	2.24	1.33	4.8	**1FL2203-2AG ■ ■ - ■ ■ C 0**	0.48	0.18	0.21	30×	0.76	1.11
6500	0.40 (0.54)	1.27	1.27	4.45	2.4	8.7	**1FL2203-4AG ■ ■ - ■ ■ C 0**	0.53	0.35	0.38		1.11	1.46
Shaft height 30 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6500	0.20 (0.27)	0.64	0.64	2.24	1.3	4.8	**1FL2203-2AF ■ ■ - ■ ■ C 0**	0.49	0.18	0.21	30×	0.77	1.12
6500	0.40 (0.54)	1.27	1.27	4.45	1.3	4.8	**1FL2203-4AF ■ ■ - ■ ■ C 0**	0.98	0.35	0.38		1.09	1.44
Shaft height 40 or 40 Compact – Rated speed n_N 3000 r/min – Line voltage 200 ... 240 V 1 AC / 3 AC													
6500	0.75 (1.02)	2.4	2.4	7.5	4.4	14.2	**1FL2204-2AG ■ ■ - ■ ■ C 0**	0.55	1.23	1.31	15×	2.6	3.4
6500	0.75 (1.02)	2.4	2.4	8.4	4.4	16	**1FL2204-2BG ■ 1 - ■ ■ C 0**	0.55	1.27	1.37		2.0	2.8
6500	1.00 (1.36)	3.2	3.2	11.3	4.4	16	**1FL2204-4AG ■ ■ - ■ ■ C 0**	0.73	1.61	1.69		3.1	3.9
6500	1.00 (1.36)	3.2	3.2	11.3	4.4	16	**1FL2204-4BG ■ 1 - ■ ■ C 0**	0.73	1.67	1.77		2.4	3.2
Shaft height 40 or 40 Compact – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6500	0.75 (1.02)	2.4	2.4	7.5	2.25	7.1	**1FL2204-2AF ■ ■ - ■ ■ C 0**	1.07	1.23	1.31	15×	2.6	3.4
6500	0.75 (1.02)	2.4	2.4	8.4	2.5	9.3	**1FL2204-2BF ■ 1 - ■ ■ C 0**	0.96	1.27	1.37		2.0	2.8
6500	1.00 (1.36)	3.2	3.2	10	3	9.9	**1FL2204-4AF ■ ■ - ■ ■ C 0**	1.07	1.61	1.69		3.1	3.9
6500	1.00 (1.36)	3.2	3.2	11.2	3	10.9	**1FL2204-4BF ■ 1 - ■ ■ C 0**	1.07	1.67	1.77		2.4	3.2
Shaft height 48 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6000	0.94 (1.36)	3.6	3	11.5	2.5	9.5	**1FL2205-2AF ■ 1 - ■ ■ B 0**	1.24	3.15	4.03	15×	3.6	4.7
6000	1.45 (2.04)	6	4.6	19.3	3.75	15.1	**1FL2205-4AF ■ 1 - ■ ■ B 0**	1.28	5.1	5.98		5	6.1

Brake	
Without	0
Holding brake	1
Degree of protection	
IP54	0
IP65 (with shaft sealing ring)	1
Shaft end, feather key	
Plain shaft	0
Shaft with feather key (half-key balancing)	1
Encoder	
Absolute encoder 17-bit singleturn	H
Absolute encoder 21-bit singleturn	S
Absolute encoder 21-bit singleturn + 12-bit multiturn	M
Connection technology	
Metal circular connector	B
Metal hybrid connector	C

In addition, the Siemens Product Configurator can be used on the internet: www.siemens.com/sinamics-s200/configuration

[1] Rated torque, rated power and maximum torque listed in the table above allow for a production tolerance of 10 %.

[2] Motor weight with absolute encoder 17-bit singleturn.

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	SINAMICS S200 servo converters Further info in section "SINAMICS S200 servo converters".			Line filter			Recommended standard fuse More information at www.siemens.com/sinamics-s200/ocpd				Recommended circuit breaker
	Max. motor power		Frame size	With one of the recommended line filters, IEC 61800-3 category C2 can be reached in combination with SINAMICS S200 for motor cable lengths up to 30 m (98.5 ft).			IEC-compliant		UL/cUL-compliant		IEC/UL/cUL-compliant
				Line voltage	I_N		I	3NA3	I	Fuse type	3RV2
	kW (hp)	Article No.		V	A	Article No.	A	Article No.	A	Class	Article No.
	SINAMICS S200 and SINAMICS S200 Basic servo converters										
	Line voltage 200 ... 240 V 1 AC / 3 AC										
	0.20 (0.27)	**6SL5 ■ 10-1BB10-2A ■ 0**	FSA	1 AC 200 ... 240	18	**6SL5550-0XB21-8CA0**	4	**3NA3804**	3	J	**3RV2011-1EA10**
				3 AC 200 ... 240	5	**6SL3203-0BE15-0VA0**					
	0.40 (0.54)	**6SL5 ■ 10-1BB10-4A ■ 0**	FSB	1 AC 200 ... 240	18	**6SL5550-0XB21-8CA0**	6	**3NA3801**	6	J	**3RV2011-1HA10**
				3 AC 200 ... 240	5	**6SL3203-0BE15-0VA0**					**3RV2011-1EA10**
	Line voltage 380 ... 480 V 3 AC										
	0.20 (0.27)	**6SL5 5 10-1BE10-2A ■ 0**	FSA	3 AC 380 ... 480	5	**6SL5550-0XE20-5CA0**	4	**3NA3804**	3	J	**3RV2011-1CA10**
	0.40 (0.54)	**6SL5 5 10-1BE10-4A ■ 0**									**3RV2011-1DA10**
	Line voltage 200 ... 240 V 1 AC / 3 AC										
	0.75 (1.02)	**6SL5 ■ 10-1BB10-8A ■ 0**	FSC	1 AC 200 ... 240	18	**6SL5550-0XB21-8CA0**	16	**3NA3805**	15	J	**3RV2011-1KA10**
				3 AC 200 ... 240	5	**6SL3203-0BE15-0VA0**					**3RV2011-1HA10**
	1.00 (1.36)	**6SL5 ■ 10-1BB11-0A■ 0**	FSC	1 AC 200 ... 240	18	**6SL5550-0XB21-8CA0**	16	**3NA3805**	15	J	**3RV2011-1KA10**
				3 AC 200 ... 240	5	**6SL3203-0BE15-0VA0**					**3RV2011-1JA10**
	Line voltage 380 ... 480 V 3 AC										
	0.75 (1.02)	**6SL5 5 10-1BE10-8A ■ 0**	FSB	3 AC 380 ... 480	5	**6SL5550-0XE20-5CA0**	10	**3NA3803**	10	J	**3RV2011-1EA10**
	1.00 (1.36)	**6SL5 5 10-1BE11-0A ■ 0**									**3RV2011-1FA10**
	Line voltage 380 ... 480 V 3 AC										
	1.00 (1.36)	**6SL5 5 10-1BE11-0A ■ 0**	FSB	3 AC 380 ... 480	5	**6SL3203-0BE15-0VA0**	10	**3NA3803**	10	J	**3RV2011-1FA10**
	1.75 (2.38)	**6SL5 5 10-1BE11-5A ■ 0**	FSC	3 AC 380 ... 480	12	**6SL5550-0XE21-2CA0**	16	**3NA3805**	15		**3RV2011-4AA10**

Servo converter	
SINAMICS S200	**5**
SINAMICS S200 Basic	**6**
Version	
Pulse Train Input (PTI)	**A**
PROFINET (PN)	**F**

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6SL3255-0AA00-5AA0

Selection and ordering data

SIMOTICS S-1FL2 servomotors – Further info in section "SIMOTICS S-1FL2 servomotors".

Max. speed	Rated power [1]	Static torque	Rated torque [1]	Max. torque [1]	Rated current	Max. current		Torque constant	Moment of inertia of rotor		Recommended load to motor inertia ratio, max.	Weight [2]	
$n_{max.}$	P_N at ΔT=100 K	M_0 at ΔT=100 K	M_N at ΔT=100 K	M_{max}	I_N at ΔT=100 K	I_{max}			$J_{without\ brake}$	$J_{with\ brake}$		$m_{without\ brake}$	$m_{with\ brake}$
r/min	kW (hp)	Nm	Nm	Nm	A	A	Article No.	Nm/A	kg cm^2	kg cm^2		kg	kg
SIMOTICS S-1FL2 Medium Inertia servomotors – For dynamic and smooth operational performance													
Shaft height 30 – Rated speed n_N 3000 r/min – Line voltage 200 ... 240 V 1 AC / 3 AC													
6500	0.20 (0.27)	0.64	0.64	2.24	1.33	4.8	**1FL2203-2AG ■ ■ - ■ ■ C 0**	0.48	0.18	0.21	30×	0.76	1.11
6500	0.40 (0.54)	1.27	1.27	4.45	2.4	8.7	**1FL2203-4AG ■ ■ - ■ ■ C 0**	0.53	0.35	0.38		1.11	1.46
Shaft height 30 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6500	0.20 (0.27)	0.64	0.64	2.24	1.3	4.8	**1FL2203-2AF ■ ■ - ■ ■ C 0**	0.49	0.18	0.21	30×	0.77	1.12
6500	0.40 (0.54)	1.27	1.27	4.45	1.3	4.8	**1FL2203-4AF ■ ■ - ■ ■ C 0**	0.98	0.35	0.38		1.09	1.44
Shaft height 40 or 40 Compact – Rated speed n_N 3000 r/min – Line voltage 200 ... 240 V 1 AC / 3 AC													
6500	0.75 (1.02)	2.4	2.4	7.5	4.4	14.2	**1FL2204-2AG ■ ■ - ■ ■ C 0**	0.55	1.23	1.31	15×	2.6	3.4
6500	0.75 (1.02)	2.4	2.4	8.4	4.4	16	**1FL2204-2BG ■ 1 - ■ ■ C 0**	0.55	1.27	1.37		2.0	2.8
6500	1.00 (1.36)	3.2	3.2	11.3	4.4	16	**1FL2204-4AG ■ ■ - ■ ■ C 0**	0.73	1.61	1.69		3.1	3.9
6500	1.00 (1.36)	3.2	3.2	11.3	4.4	16	**1FL2204-4BG ■ 1 - ■ ■ C 0**	0.73	1.67	1.77		2.4	3.2
Shaft height 40 or 40 Compact – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6500	0.75 (1.02)	2.4	2.4	7.5	2.25	7.1	**1FL2204-2AF ■ ■ - ■ ■ C 0**	1.07	1.23	1.31	15×	2.6	3.4
6500	0.75 (1.02)	2.4	2.4	8.4	2.5	9.3	**1FL2204-2BF ■ 1 - ■ ■ C 0**	0.96	1.27	1.37		2.0	2.8
6500	1.00 (1.36)	3.2	3.2	10	3	9.9	**1FL2204-4AF ■ ■ - ■ ■ C 0**	1.07	1.61	1.69		3.1	3.9
6500	1.00 (1.36)	3.2	3.2	11.2	3	10.9	**1FL2204-4BF ■ 1 - ■ ■ C 0**	1.07	1.67	1.77		2.4	3.2
Shaft height 48 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6000	0.94 (1.36)	3.6	3	11.5	2.5	9.5	**1FL2205-2AF ■ 1 - ■ ■ B 0**	1.24	3.15	4.03	15×	3.6	4.7
6000	1.45 (2.04)	6	4.6	19.3	3.75	15.1	**1FL2205-4AF ■ 1 - ■ ■ B 0**	1.28	5.1	5.98		5	6.1

Brake					
Without	**0**				
Holding brake	**1**				
Degree of protection					
IP54		**0**			
IP65 (with shaft sealing ring)		**1**			
Shaft end, feather key					
Plain shaft			**0**		
Shaft with feather key (half-key balancing)			**1**		
Encoder					
Absolute encoder 17-bit singleturn				**H**	
Absolute encoder 21-bit singleturn				**S**	
Absolute encoder 21-bit singleturn + 12-bit multiturn				**M**	
Connection technology					
Metal circular connector					**B**
Metal hybrid connector					**C**

In addition, the Siemens Product Configurator can be used on the internet: www.siemens.com/sinamics-s200/configuration

[1] Rated torque, rated power and maximum torque listed in the table above allow for a production tolerance of 10 %.

[2] Motor weight with absolute encoder 17-bit singleturn.

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	MOTION-CONNECT connection systems – Further info in section "MOTION-CONNECT 350 and 380 connection systems".					
	Pre-assembled dual cables for power and communication		Pre-assembled power cables		Pre-assembled cables for communication	
					SINAMICS S200 – Absolute encoder on the 1FL2 servomotor	
	No. of cores × cross-section mm²	Article No.	No. of cores × cross-section mm²	Article No.	No. of cores × cross-section mm²	Article No.
	MOTION-CONNECT connection systems					
	4 × 0.75 / 3 × 2 × 0.25 [1] or 4 × 0.75 / 3 × 2 × 0.25 [1] / 2 × 0.38	**6FX3 ■02-7 ■ ■01-1 ■ ■0**	–	–	–	–
	4 × 0.75 / 3 × 2 × 0.25 [1] or 4 × 0.75 / 3 × 2 × 0.25 [1] / 2 × 0.38	**6FX3 ■02-7 ■ ■01-1 ■ ■0**	–	–	–	–
	4 × 0.75 / 3 × 2 × 0.25 [1] or 4 × 0.75 / 3 × 2 × 0.25 [1] / 2 × 0.38	**6FX3 ■02-7 ■ ■01-1 ■ ■0**	–	–	–	–
	4 × 0.75 / 3 × 2 × 0.25 [1] or 4 × 0.75 / 3 × 2 × 0.25 [1] / 2 × 0.38	**6FX3 ■02-7 ■ ■01-1 ■ ■0**	–	–	–	–
	–	–	4 × 0.75 or 4 × 0.75 / 2 × 0.38	**6FX3 ■ 02-5 ■ K32-1 ■ ■0**	3 × 2 × 0.25 [1]	**6FX3 ■02-2XB11-1 ■ ■0**
	Connection system					
	MOTION-CONNECT 350	**5**		**5**		**5**
	MOTION-CONNECT 380	**8**		**8**		**8**
	Brake cores					
	Without					
	With	**C D**		**C D**		
	Outlet direction					
	Drive end (DE)					
	Non drive end (NDE)	**D N**				
	Length					
	3 m	**A A A B C D**	**D F H A F A A**			**A A A B C D**	**D F H A F A A**
	5 m						
	7 m						
	10 m						
	15 m						
	20 m						
	30 m			**A A A B C D**	**D F H A F A A**		

[1] No. of cores × cross-section for encoder cable in mm² for:
Length <20 m: 3 x 2 x 0.25
Length ≥ 20 m: 2 x 2 x 0.25 + 2 x 0.38

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6SL3255-0AA00-5AA0



SINAMICS S200 servo drive system with SIMOTICS S-1FL2 Medium Inertia servomotors

Selection and ordering data

SIMOTICS S-1FL2 servomotors – Further info in section "SIMOTICS S-1FL2 servomotors".

Max. speed	Rated power [1]	Static torque	Rated torque [1]	Max. torque [1]	Rated current	Max. current		Torque constant	Moment of inertia of rotor		Recommended load to motor inertia ratio, max.	Weight [2]	
$n_{max.}$	P_N at ΔT=100 K	M_0 at ΔT=100 K	M_N at ΔT=100 K	M_{max}	I_N at ΔT=100 K	I_{max}			$J_{without\ brake}$	$J_{with\ brake}$		$m_{without\ brake}$	$m_{with\ brake}$
r/min	kW (hp)	Nm	Nm	Nm	A	A	Article No.	Nm/A	kg cm^2	kg cm^2		kg	kg
SIMOTICS S-1FL2 Medium Inertia servomotors – For dynamic and smooth operational performance													
Shaft height 30 – Rated speed n_N 3000 r/min – Line voltage 200 ... 240 V 1 AC / 3 AC													
6500	0.20 (0.27)	0.64	0.64	2.24	1.33	4.8	**1FL2203-2AG ■ ■ - ■ ■ C 0**	0.48	0.18	0.21	30×	0.76	1.11
6500	0.40 (0.54)	1.27	1.27	4.45	2.4	8.7	**1FL2203-4AG ■ ■ - ■ ■ C 0**	0.53	0.35	0.38		1.11	1.46
Shaft height 30 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6500	0.20 (0.27)	0.64	0.64	2.24	1.3	4.8	**1FL2203-2AF ■ ■ - ■ ■ C 0**	0.49	0.18	0.21	30×	0.77	1.12
6500	0.40 (0.54)	1.27	1.27	4.45	1.3	4.8	**1FL2203-4AF ■ ■ - ■ ■ C 0**	0.98	0.35	0.38		1.09	1.44
Shaft height 40 or 40 Compact – Rated speed n_N 3000 r/min – Line voltage 200 ... 240 V 1 AC / 3 AC													
6500	0.75 (1.02)	2.4	2.4	7.5	4.4	14.2	**1FL2204-2AG ■ ■ - ■ ■ C 0**	0.55	1.23	1.31	15×	2.6	3.4
6500	0.75 (1.02)	2.4	2.4	8.4	4.4	16	**1FL2204-2BG ■ 1 - ■ ■ C 0**	0.55	1.27	1.37		2.0	2.8
6500	1.00 (1.36)	3.2	3.2	11.3	4.4	16	**1FL2204-4AG ■ ■ - ■ ■ C 0**	0.73	1.61	1.69		3.1	3.9
6500	1.00 (1.36)	3.2	3.2	11.3	4.4	16	**1FL2204-4BG ■ 1 - ■ ■ C 0**	0.73	1.67	1.77		2.4	3.2
Shaft height 40 or 40 Compact – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6500	0.75 (1.02)	2.4	2.4	7.5	2.25	7.1	**1FL2204-2AF ■ ■ - ■ ■ C 0**	1.07	1.23	1.31	15×	2.6	3.4
6500	0.75 (1.02)	2.4	2.4	8.4	2.5	9.3	**1FL2204-2BF ■ 1 - ■ ■ C 0**	0.96	1.27	1.37		2.0	2.8
6500	1.00 (1.36)	3.2	3.2	10	3	9.9	**1FL2204-4AF ■ ■ - ■ ■ C 0**	1.07	1.61	1.69		3.1	3.9
6500	1.00 (1.36)	3.2	3.2	11.2	3	10.9	**1FL2204-4BF ■ 1 - ■ ■ C 0**	1.07	1.67	1.77		2.4	3.2
Shaft height 48 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
6000	0.94 (1.36)	3.6	3	11.5	2.5	9.5	**1FL2205-2AF ■ 1 - ■ ■ B 0**	1.24	3.15	4.03	15×	3.6	4.7
6000	1.45 (2.04)	6	4.6	19.3	3.75	15.1	**1FL2205-4AF ■ 1 - ■ ■ B 0**	1.28	5.1	5.98		5	6.1

Brake	
Without	0
Holding brake	1
Degree of protection	
IP54	0
IP65 (with shaft sealing ring)	1
Shaft end, feather key	
Plain shaft	0
Shaft with feather key (half-key balancing)	1
Encoder	
Absolute encoder 17-bit singleturn	H
Absolute encoder 21-bit singleturn	S
Absolute encoder 21-bit singleturn + 12-bit multiturn	M
Connection technology	
Metal circular connector	B
Metal hybrid connector	C

In addition, the Siemens Product Configurator can be used on the internet: www.siemens.com/sinamics-s200/configuration

[1] Rated torque, rated power and maximum torque listed in the table above allow for a production tolerance of 10 %.

[2] Motor weight with absolute encoder 17-bit singleturn.

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MOTION-CONNECT connection systems (continued) – Further info in section "MOTION-CONNECT 350 and 380 connection systems".

	Pre-assembled cables for communication		**Connectors**				
	SINAMICS S200 – Brake on the 1FL2 servomotor with holding brake		Drive end (DE) direction	Non drive end (NDE) direction	Motor side		
	No. of cores × cross-section mm²	Article No.	for hybrid connection Article No.	for hybrid connection Article No.	for power connection Article No.	for encoder connection Article No.	for brake Article No.
	MOTION-CONNECT connection systems (continued)						
	–	–	**6FX2003-3HD00**	**6FX2003-3HN00**	–	–	–
	–	–	**6FX2003-3HD00**	**6FX2003-3HN00**	–	–	–
	–	–	**6FX2003-3HD00**	**6FX2003-3HN00**	–	–	–
	–	–	**6FX2003-3HD00**	**6FX2003-3HN00**	–	–	–
	–	–	–	–	**6FX2003-0LL63**	**6FX2003-0DB63**	– (in power connector)

MOTION-CONNECT connectors for SINAMICS S200 servo converters				
Crimp contacts "power and PE" (2000 pieces per reel) for hybrid connection	Crimp contacts "signal and brake" (2000 pieces per reel) for hybrid connection	Connectors converter side		
		for power connection	IX signal connector, C coded for encoder connection	for brake
6FX2003-7HP00	**6FX2003-7HS00**	–	**6FX2003-0DE61**	–

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6SL3255-0AA00-5AA0



SINAMICS S200 servo drive system with SIMOTICS S-1FL2 High Inertia servomotors

Selection and ordering data

SIMOTICS S-1FL2 servomotors – Further info in section "SIMOTICS S-1FL2 servomotors".

Max. speed	Rated power [1]	Static torque	Rated torque [1]	Max. torque [1]	Rated current	Max. current		Torque constant	Moment of inertia of rotor		Recommended load to motor inertia ratio, max.	Weight [2]	
$n_{max.}$	P_N at ΔT=100 K	M_0 at ΔT=100 K	M_N at ΔT=100 K	M_{max}	I_N at ΔT=100 K	I_{max}			$J_{without\ brake}$	$J_{with\ brake}$		$m_{without\ brake}$	$m_{with\ brake}$
r/min	kW (hp)	Nm	Nm	Nm	A	A	Article No.	Nm/A	kg cm^2	kg cm^2		kg	kg
SIMOTICS S-1FL2 High Inertia servomotors – For smooth operational performance													
Shaft height 45 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
4000	0.40 (0.54)	1.45	1.27	4.445	1.2	4.2	**1FL2304-2AF ■ 1 - ■ ■ B 0**	1.12	2.69	3.05	10×	3.5	4.9
4000	0.75 (1.02)	3.02	2.39	8.365	2.1	7.3	**1FL2304-4AF ■ 1 - ■ ■ B 0**	1.21	5.18	5.54		5.4	6.8
Shaft height 65 – Rated speed n_N 2000 r/min – Line voltage 380 ... 480 V 3 AC													
3000	0.75 (1.02)	3.7	3.58	12	2.5	8.4	**1FL2306-1AC ■ 1 - ■ ■ B 0**	1.48	7.962	9.57	10×	5.9	8.9
3000	1.00 (1.36)	5	4.78	17	3	10.7	**1FL2306-2AC ■ 1 - ■ ■ B 0**	1.67	11.65	13.26	5×	7.3	10.3
3000	1.50 (2.04)	8	7.16	22	4.6	14.1	**1FL2306-4AC ■ 1 - ■ ■ B 0**	1.63	15.287	16.9		8.6	11.6
3000	1.75 (2.38)	8.8	8.4	30	5.3	19	**1FL2306-6AC ■ 1 - ■ ■ B 0**	1.66	22.612	24.22		11.1	14.1
3000	2.00 (2.72)	13.3	9.6	33.425	5.9	20.7	**1FL2306-7AC ■ 1 - ■ ■ B 0**	1.71	29.938	31.55		13.7	16.7
Shaft height 90 – Rated speed n_N 2000 r/min – Line voltage 380 ... 480 V 3 AC													
3000	2.50 (3.40)	12.6	11.9	37.5	7.8	25.2	**1FL2310-0AC ■ 1 - ■ ■ B 0**	1.62	47.409	56.78	5×	15.4	21.4
3000	3.50 (4.76)	18.1	16.7	55	11	36.2	**1FL2310-2AC ■ 1 - ■ ■ B 0**	1.65	69.095	78.47		19.4	25.4
2500	5.00 (6.80)	25.2	23.9	70	12.6	36.9	**1FL2310-4AC ■ 1 - ■ ■ B 0**	2	90.827	100.2		23.7	29.7
2000	7.00 (9.52)	35.2	33.4	90	13.2	35.6	**1FL2310-6AC ■ 1 - ■ ■ B 0**	2.67	134.292	143.66		31.9	37.9

Brake Without Holding brake	**0** **1**				
Degree of protection IP54 IP65 (with shaft sealing ring)		**0** **1**			
Shaft end, feather key Plain shaft Shaft with feather key (half-key balancing)			**0** **1**		
Encoder Absolute encoder 17-bit singleturn Absolute encoder 21-bit singleturn Absolute encoder 21-bit singleturn + 12-bit multiturn				**H** **S** **M**	
Connection technology Metal circular connector Metal hybrid connector					**B** **C**

In addition, the Siemens Product Configurator can be used on the internet: www.siemens.com/sinamics-s200/configuration

[1] Rated torque, rated power and maximum torque listed in the table above allow for a production tolerance of 10 %.

[2] Motor weight with absolute encoder 17-bit singleturn.

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	SINAMICS S200 servo converters Further info in section "SINAMICS S200 servo converters".			Line filter			Recommended standard fuse More information at www.siemens.com/sinamics-s200/ocpd				Recommended circuit breaker
	Max. motor power		Frame size	With one of the recommended line filters, IEC 61800-3 category C2 can be reached in combination with SINAMICS S200 for motor cable lengths up to 30 m (98.5 ft).			IEC-compliant		UL/cUL-compliant		IEC/UL/cUL-compliant
				Line voltage	I_N		I	3NA3	I	Fuse type	3RV2
	kW (hp)	Article No.		V	A	Article No.	A	Article No.	A	Class	Article No.
	SINAMICS S200 servo converters										
	Line voltage 380 ... 480 V 3 AC										
	0.40 (0.54)	**6SL5 5 10-1BE10-4A ■ 0**	FSA	3 AC 380 ... 480	5	**6SL5550-0XE20-5CA0**	4	**3NA3804**	3	J	**3RV2011-1DA10**
	0.75 (1.02)	**6SL5 5 10-1BE10-8A ■ 0**	FSB				10	**3NA3803**	10		**3RV2011-1EA10**
	Line voltage 380 ... 480 V 3 AC										
	0.75 (1.02)	**6SL5 5 10-1BE10-8A ■ 0**	FSB	3 AC 380 ... 480	5	**6SL5550-0XE20-5CA0**	10	**3NA3803**	10	J	**3RV2011-1EA10**
	1.00 (1.36)	**6SL5 5 10-1BE11-0A ■ 0**									**3RV2011-1FA10**
	1.75 (2.38)	**6SL5 5 10-1BE11-5A ■ 0**	FSC	3 AC 380 ... 480	12	**6SL5550-0XE21-2CA0**	16	**3NA3805**	15		**3RV2011-4AA10**
	2.50 (3.40)	**6SL5 5 10-1BE12-5A ■ 0**									**3RV2011-4AA10**
	Line voltage 380 ... 480 V 3 AC										
	2.50 (3.40)	**6SL5 5 10-1BE12-5A ■ 0**	FSC	3 AC 380 ... 480	12	**6SL5550-0XE21-2CA0**	16	**3NA3805**	20	J	**3RV2011-4AA10**
	3.50 (4.76)	**6SL5 5 10-1BE13-5A ■ 0**	FSD	3 AC 380 ... 480	20	**6SL5550-0XE22-0CA0**	20	**3NA3807**	25		**3RV2021-4BA10**
	5.00 (6.80)	**6SL5 5 10-1BE15-0A ■ 0**									**3RV2021-4BA10**
	7.00 (9.52)	**6SL5 5 10-1BE17-0A ■ 0**					25	**3NA3810**			**3RV2021-4DA10**

Servo converter	
SINAMICS S200	**5**
SINAMICS S200 Basic	**6**
Version	
Pulse Train Input (PTI)	**A**
PROFINET (PN)	**F**

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Selection and ordering data

SIMOTICS S-1FL2 servomotors – Further info in section "SIMOTICS S-1FL2 servomotors".

Max. speed	Rated power [1]	Static torque	Rated torque [1]	Max. torque [1]	Rated current	Max. current		Torque constant	Moment of inertia of rotor		Recommended load to motor inertia ratio, max.	Weight [2]	
$n_{max.}$	P_N at ΔT=100 K	M_0 at ΔT=100 K	M_N at ΔT=100 K	M_{max}	I_N at ΔT=100 K	I_{max}			$J_{without\ brake}$	$J_{with\ brake}$		$m_{without\ brake}$	$m_{with\ brake}$
r/min	kW (hp)	Nm	Nm	Nm	A	A	Article No.	Nm/A	kg cm^2	kg cm^2		kg	kg
SIMOTICS S-1FL2 High Inertia servomotors – For smooth operational performance													
Shaft height 45 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
4000	0.40 (0.54)	1.45	1.27	4.445	1.2	4.2	**1FL2304-2AF ■ 1 - ■ ■ B 0**	1.12	2.69	3.05	10×	3.5	4.9
4000	0.75 (1.02)	3.02	2.39	8.365	2.1	7.3	**1FL2304-4AF ■ 1 - ■ ■ B 0**	1.21	5.18	5.54		5.4	6.8
Shaft height 65 – Rated speed n_N 2000 r/min – Line voltage 380 ... 480 V 3 AC													
3000	0.75 (1.02)	3.7	3.58	12	2.5	8.4	**1FL2306-1AC ■ 1 - ■ ■ B 0**	1.48	7.962	9.57	10×	5.9	8.9
3000	1.00(1.36)	5	4.78	17	3	10.7	**1FL2306-2AC ■ 1 - ■ ■ B 0**	1.67	11.65	13.26	5×	7.3	10.3
3000	1.50 (2.04)	8	7.16	22	4.6	14.1	**1FL2306-4AC ■ 1 - ■ ■ B 0**	1.63	15.287	16.9		8.6	11.6
3000	1.75 (2.38)	8.8	8.4	30	5.3	19	**1FL2306-6AC ■ 1 - ■ ■ B 0**	1.66	22.612	24.22		11.1	14.1
3000	2.00 (2.72)	13.3	9.6	33.425	5.9	20.7	**1FL2306-7AC ■ 1 - ■ ■ B 0**	1.71	29.938	31.55		13.7	16.7
Shaft height 90 – Rated speed n_N 2000 r/min – Line voltage 380 ... 480 V 3 AC													
3000	2.50 (3.40)	12.6	11.9	37.5	7.8	25.2	**1FL2310-0AC ■ 1 - ■ ■ B 0**	1.62	47.409	56.78	5×	15.4	21.4
3000	3.50 (4.76)	18.1	16.7	55	11	36.2	**1FL2310-2AC ■ 1 - ■ ■ B 0**	1.65	69.095	78.47		19.4	25.4
2500	5.00 (6.80)	25.2	23.9	70	12.6	36.9	**1FL2310-4AC ■ 1 - ■ ■ B 0**	2	90.827	100.2		23.7	29.7
2000	7.00 (9.52)	35.2	33.4	90	13.2	35.6	**1FL2310-6AC ■ 1 - ■ ■ B 0**	2.67	134.292	143.66		31.9	37.9

Brake Without Holding brake	**0** **1**				
Degree of protection IP54 IP65 (with shaft sealing ring)		**0** **1**			
Shaft end, feather key Plain shaft Shaft with feather key (half-key balancing)			**0** **1**		
Encoder Absolute encoder 17-bit singleturn Absolute encoder 21-bit singleturn Absolute encoder 21-bit singleturn + 12-bit multiturn				**H** **S** **M**	
Connection technology Metal circular connector Metal hybrid connector					**B** **C**

In addition, the Siemens Product Configurator can be used on the internet: www.siemens.com/sinamics-s200/configuration

[1] Rated torque, rated power and maximum torque listed in the table above allow for a production tolerance of 10 %.

[2] Motor weight with absolute encoder 17-bit singleturn.

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	MOTION-CONNECT connection systems – Further info in section "MOTION-CONNECT 350 and 380 connection systems".					
	Pre-assembled dual cables for power and communication		**Pre-assembled power cables**		**Pre-assembled cables for communication**	
					SINAMICS S200 – Absolute encoder on the 1FL2 servomotor	
	No. of cores × cross-section mm²	Article No.	No. of cores × cross-section mm²	Article No.	No. of cores × cross-section mm²	Article No.
	MOTION-CONNECT connection systems					
	–	–	4 × 0.75	**6FX3 ■ 02-5 C L01-1 ■ ■0**	3 × 2 × 0.25 [1]	**6FX3 ■02-2XB12-1 ■ ■0**
	–	–	4 × 0.75	**6FX3 ■ 02-5 C L01-1 ■ ■0**	3 × 2 × 0.25 [1]	**6FX3 ■02-2XB12-1 ■ ■0**
	–	–	4 × 1.5	**6FX3 ■ 02-5 C L02-1 ■ ■0**	3 × 2 × 0.25 [1]	**6FX3 ■02-2XB12-1 ■ ■0**
			4 × 2.5	**6FX3 ■ 02-5 C L12-1 ■ ■0**		

Connection system																								
MOTION-CONNECT 350																								
MOTION-CONNECT 380																								
								5												**5**				
								8												**8**				
Brake cores																								
Without										**C**														
With										**D**														
Length																								
3 m											**A**	**D**											**A**	**D**
5 m											**A**	**F**											**A**	**F**
7 m											**A**	**H**											**A**	**H**
10 m											**B**	**A**											**B**	**A**
15 m											**B**	**F**											**C**	**F**
20 m											**C**	**A**											**D**	**A**
30 m											**D**	**A**												

[1] No. of cores × cross-section for encoder cable in mm² for:
Length <20 m: 3 x 2 x 0.25
Length ≥ 20 m: 2 x 2 x 0.25 + 2 x 0.38

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Selection and ordering data

SIMOTICS S-1FL2 servomotors – Further info in section "SIMOTICS S-1FL2 servomotors".

Max. speed	Rated power [1]	Static torque	Rated torque [1]	Max. torque [1]	Rated current	Max. current		Torque constant	Moment of inertia of rotor		Recommended load to motor inertia ratio, max.	Weight [2]	
$n_{max.}$	P_N at ΔT=100 K	M_0 at ΔT=100 K	M_N at ΔT=100 K	M_{max}	I_N at ΔT=100 K	I_{max}			$J_{without\ brake}$	$J_{with\ brake}$		$m_{without\ brake}$	$m_{with\ brake}$
r/min	kW (hp)	Nm	Nm	Nm	A	A	Article No.	Nm/A	kg cm^2	kg cm^2		kg	kg
SIMOTICS S-1FL2 High Inertia servomotors – For smooth operational performance													
Shaft height 45 – Rated speed n_N 3000 r/min – Line voltage 380 ... 480 V 3 AC													
4000	0.40 (0.54)	1.45	1.27	4.445	1.2	4.2	**1FL2304-2AF ■ 1 - ■ ■ B 0**	1.12	2.69	3.05	10×	3.5	4.9
4000	0.75 (1.02)	3.02	2.39	8.365	2.1	7.3	**1FL2304-4AF ■ 1 - ■ ■ B 0**	1.21	5.18	5.54		5.4	6.8
Shaft height 65 – Rated speed n_N 2000 r/min – Line voltage 380 ... 480 V 3 AC													
3000	0.75 (1.02)	3.7	3.58	12	2.5	8.4	**1FL2306-1AC ■ 1 - ■ ■ B 0**	1.48	7.962	9.57	10×	5.9	8.9
3000	1.00 (1.36)	5	4.78	17	3	10.7	**1FL2306-2AC ■ 1 - ■ ■ B 0**	1.67	11.65	13.26	5×	7.3	10.3
3000	1.50 (2.04)	8	7.16	22	4.6	14.1	**1FL2306-4AC ■ 1 - ■ ■ B 0**	1.63	15.287	16.9		8.6	11.6
3000	1.75 (2.38)	8.8	8.4	30	5.3	19	**1FL2306-6AC ■ 1 - ■ ■ B 0**	1.66	22.612	24.22		11.1	14.1
3000	2 (2.72)	13.3	9.6	33.425	5.9	20.7	**1FL2306-7AC ■ 1 - ■ ■ B 0**	1.71	29.938	31.55		13.7	16.7
Shaft height 90 – Rated speed n_N 2000 r/min – Line voltage 380 ... 480 V 3 AC													
3000	2.50 (3.40)	12.6	11.9	37.5	7.8	25.2	**1FL2310-0AC ■ 1 - ■ ■ B 0**	1.62	47.409	56.78	5×	15.4	21.4
3000	3.50 (4.76)	18.1	16.7	55	11	36.2	**1FL2310-2AC ■ 1 - ■ ■ B 0**	1.65	69.095	78.47		19.4	25.4
2500	5.00 (6.80)	25.2	23.9	70	12.6	36.9	**1FL2310-4AC ■ 1 - ■ ■ B 0**	2	90.827	100.2		23.7	29.7
2000	7.00 (9.52)	35.2	33.4	90	13.2	35.6	**1FL2310-6AC ■ 1 - ■ ■ B 0**	2.67	134.292	143.66		31.9	37.9

Brake Without Holding brake	**0** **1**				
Degree of protection IP54 IP65 (with shaft sealing ring)		**0** **1**			
Shaft end, feather key Plain shaft Shaft with feather key (half-key balancing)			**0** **1**		
Encoder Absolute encoder 17-bit singleturn Absolute encoder 21-bit singleturn Absolute encoder 21-bit singleturn + 12-bit multiturn				**H** **S** **M**	
Connection technology Metal circular connector Metal hybrid connector					**B** **C**

In addition, the Siemens Product Configurator can be used on the internet: www.siemens.com/sinamics-s200/configuration

[1] Rated torque, rated power and maximum torque listed in the table above allow for a production tolerance of 10 %.

[2] Motor weight with absolute encoder 17-bit singleturn.

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MOTION-CONNECT connection systems (continued) – Further info in section "MOTION-CONNECT 350 and 380 connection systems".

Pre-assembled cables for communication		Connectors				
SINAMICS S200 – Brake on the 1FL2 servomotor with holding brake		Drive end (DE) direction	Non drive end (NDE) direction	Motor side		
No. of cores × cross-section mm²	Article No.	for hybrid connection Article No.	for hybrid connection Article No.	for power connection Article No.	for encoder connection Article No.	for brake Article No.
MOTION-CONNECT connection systems (continued)						
2 × 0.75	**6FX3 ■02-5BL03-1 ■ ■ 0**	–	–	**6FX2003-0LL13**	**6FX2003-0DB13**	**6FX2003-0LL53**
2 × 0.75	**6FX3 ■02-5BL03-1 ■ ■ 0**	–	–	**6FX2003-0LL13**	**6FX2003-0DB13**	**6FX2003-0LL53**
2 × 0.75	**6FX3 ■02-5BL03-1 ■ ■ 0**	–	–	**6FX2003-0LL13**	**6FX2003-0DB13**	**6FX2003-0LL53**

Connection system	
MOTION-CONNECT 350	**5**
MOTION-CONNECT 380	**8**

Length		
3 m	**A**	**D**
5 m	**A**	**F**
7 m	**A**	**H**
10 m	**B**	**A**
15 m	**B**	**F**
20 m	**C**	**A**
30 m	**D**	**A**

MOTION-CONNECT connectors for SINAMICS S200 servo converters

Connectors converter side		
for power connection	IX signal connector, C coded for encoder connection	for brake
–	**6FX2003-0DE61**	–

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Accessories

Accessories for SINAMICS S200 Basic and SINAMICS S200 servo converters

Description	Article No.
SINAMICS SD card 8 GB Parameter assignments, firmware and licenses for a converter can be stored on the memory card.	
• Empty	**6SL5970-0AA00-0AA0**
• With firmware V6.2	**6SL5570-0GC00-0AA0**
• With firmware V6.3	**6SL5570-0GD00-0AA0**
• With firmware V6.4	**6SL5570-0GE00-0AA0**
• With firmware V6.4 and SINAMICS OPC UA Single Extended license [1]	**6SL5570-0GE00-0AA0-Z H80**
SINAMICS UPC UA Single Extended license for SINAMICS S200 [1] • Certificate of License (CoL) in electronic form	**6SL5977-0AA00-1GA0**
SINAMICS SDI Pro 5.5" NEW Powerful and user-friendly operator panel for the next generation of SINAMICS converters SINAMICS S200, SINAMICS S210 (6SL5...) and SINAMICS G220	**6SL4950-0AH35-2AA0**
SINAMICS Smart Adapter Wi-Fi solution for the next generation of SINAMICS converters SINAMICS S200, SINAMICS S210 (6SL5...) and SINAMICS G220	**6SL4950-0AJ00-0AA0**
Connecting cables and connectors for SIMATIC S7 controller and SINAMICS S200 Basic / SINAMICS S200, Pulse Train Input version (PTI)	
Connector for setpoint cable 50-pin MDR connector	**6SL3260-2NA00-0VA0**
Setpoint cable with connector MDR 50-pin connector, free pins to controller side, length: 1 m	**6SL3260-4NA00-1VB0**
Setpoint cable with connectors on both sides and separate terminal block MDR 50-pin connector, terminal block to controller side, length: 0.5 m	**6SL5562-0XX00-0AD0**
Connecting cables and connectors for SIMATIC S7 controller and SINAMICS S200 Basic / SINAMICS S200, PROFINET version	
PROFINET I/O connector 20-pin	**6SL3260-2MA00-0VA0**
PROFINET I/O cable with 20-pin MDR connector, length: 1 m (3.28 ft)	**6SL5562-1XX00-0AC0**
PROFINET I/O adapter for 20-pin MDR connector	**6SL5562-1XX00-0AD0**
Pre-assembled PROFINET cable with two RJ45 180° plugs, length: 1 m (3.28 ft)	**6XV1871-5BH10**
RJ45 data plug-in connector with 180° (straight) cable outlet	**6GK1901-1BB10-2AA0**
Standard PROFINET cable 4-core, sold by the meter, not assembled	**6XV1840-2AH10**
PROFINET patch cable For the networking of concatenated converters Industrial Ethernet TP cord, CAT 6 A, twisted pair line 4 × 2 cores, pre-assembled with two RJ45 connectors	
• 0.3 m (0.98 ft)	**6XV1870-3QE30**
• 0.5 m (1.64 ft)	**6XV1870-3QE50**

For further information about PROFINET cables refer on the internet at: www.siemens.com/simatic-net

Recommended SIMATIC S7 controller

More information about SIMATIC S7-1500 and SIMATIC S7-1500 T-CPU controllers is available in Catalog ST 70 and on the internet at: www.siemens.com/simatic-s7-1500

Spare parts for SINAMICS S200 Basic and SINAMICS S200 servo converters

Description	Article No.
Shield connection kit including several shield plates and fixing screws	**6SL5566-0AB00-0AA0**
Terminal kits	
• for SINAMICS S200	**6SL5568-0XX00-0AA0**
• for SINAMICS S200 Basic	**6SL5668-0XX00-0AA0**
Replacement fans	
• for SINAMICS S200, 400 V version, FSC	**6SL5560-0CE00-0AA0**
• for SINAMICS S200, 400 V version, FSD	**6SL5560-0DE00-0AA0**

Starter kit

Description	Article No.
SINAMICS S200 starter kit **PROFINET version** Scope of delivery: • SINAMICS S200 servo converter, PROFINET version, 200 V, 400 W • SIMOTICS S-1FL2 Low Inertia servomotor, shaft height 30, 400 W, IP65 degree of protection, without holding brake, shaft with feather key, with absolute encoder 21-bit singleturn • Pre-assembled dual motor connection cable, drive end, 3 m (9.84 ft) • Pre-assembled PROFINET I/O cable, 1 m (3.28 ft) The delivery quantity is limited to three units per customer.	**6SL5990-0AE50-0AA0**

Training cases

Description	Article No.
SINAMICS S200 training case **PROFINET version** Scope of delivery: • 2 × SINAMICS S200 servo converter, PROFINET version, 200 V, 100 W • 1 × SIMOTICS S-1FL2 Low Inertia servomotor, absolute encoder 21-bit singleturn • 1 × SIMOTICS S-1FL2 Low Inertia servomotor, absolute encoder 21-bit singleturn + 12-bit multiturn • 2 × Pre-assembled dual motor connection cable • Rail, prepared for installation of a controller, e.g. SIMATIC S7-1500 (controller not included in scope of delivery) • SIMATIC ET 200SP, PROFINET Interface Module The SINAMICS S200 training case is supplied as a trolley with a hood.	**6AG1067-1AA42-0AA0**
SINAMICS S200 training case **Pulse Train Input version (PTI)** Scope of delivery: • 1 × SINAMICS S200 servo converter, Pulse Train Input version (PTI), 200 V, 100 W • 1 × SIMOTICS S-1FL2 Low Inertia servomotor, absolute encoder 21-bit singleturn + 12-bit multiturn • 1 × Pre-assembled dual motor connection cable • 1 × SIMATIC S7-1200, compact CPU, CPU 1211C, DC/DC/DC • 1 × SIMATIC S7-1200, Communication Board CB 1241, RS485 The SINAMICS S200 training case Pulse Train Input version (PTI) is supplied as a trolley with a hood.	**6AG1067-1AA43-0AA0**

[1] OPC UA server with information model for inventory management and process data acquisition will be available soon with an V6.4 HF.

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More information

Identification link according to IEC 61406 for SINAMICS S200

The ID link contains the article and serial number of the product. As a QR code, it replaces the previous data matrix code on the nameplate and takes you with the URL directly to a product information page on the internet with access to the technical documentation, data sheet, certificates, FAQs, product notifications, and catalogs. Paper package inserts become superfluous since the information is available electronically directly via the QR code, even years later. In this way, we are making a valuable contribution to the preservation of our environment. You don't need an additional app. Simply scan the QR code with your smartphone or tablet. According to IEC 61406, the QR code of an ID link is marked with a frame and a triangle at the bottom right. With their globally unique identifiers, Siemens products are ready for Industry 4.0.

The ID serves as a connection to the administration shell with which modules of the digital twin can be provided.

Further documentation, such as the operating instructions, is available free on the internet at:
www.siemens.com/sinamics-s200/documentation

Detailed information on the SINAMICS S200 drive system, including the latest technical documentation (brochures, tutorials, dimensional drawings, certificates, manuals and operating instructions), is available on the internet at:
www.siemens.com/sinamics-s200

and is also available via the Siemens Product Configurator on the internet.
The Siemens Product Configurator can be found in SiePortal at the following address:
www.siemens.com/sinamics-s200/configuration

Selection and ordering data

Description	Article No.
SINAMICS S200 starter kit **PROFINET version** Scope of delivery: • SINAMICS S200 servo converter, PROFINET version, 200 V, 400 W • SIMOTICS S-1FL2 Low Inertia servomotor, shaft height 30, 400 W, IP65 degree of protection, without holding brake, shaft with feather key, with absolute encoder 21-bit singleturn • Pre-assembled dual motor connection cable, drive end, 3 m (9.84 ft) • Pre-assembled PROFINET I/O cable, 1 m (3.28 ft) The delivery quantity is limited to three units per customer.	**6SL5990-0AE50-0AA0**

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Selection and ordering data

Recommended controller for SINAMICS S200 pulse train (PTI) version

SIMATIC S7-1200 Basic Controller for SINAMICS S200 pulse train (PTI) version							
Compact CPU expandable with			Digital outputs	of which high-speed outputs (Pulse Train Output)	Work memory	CPU	
Signal board or communication board	Communication modules	Signal Modules					Article No.
1	3	–	4	4 (100 kHz)	50 KB	CPU 1211C DC/DC/DC	**6ES7211-1AE40-0XB0**
		2	6	4 (100 kHz)	75 KB	CPU 1212C DC/DC/DC	**6ES7212-1AE40-0XB0**
		8	10	4 (100 kHz)	100 KB	CPU 1214C DC/DC/DC	**6ES7214-1AG40-0XB0**
					125 KB	CPU 1215C DC/DC/DC	**6ES7215-1AG40-0XB0**
				4 (1 MHz)	150 KB	CPU 1217C DC/DC/DC	**6ES7217-1AG40-0XB0**

One SIMATIC S7-1200 CPU can control up to 4 SINAMICS S200 axes. Each axis requires 2 high-speed digital outputs for the pulse train interface. The SIMATIC S7-1500 compact CPUs can also be connected to SINAMICS S200 via “pulse train”.

Expansion for control of more than 2 axes			
Signal boards, 0.1 A, 200 kHz, can be plugged directly into the CPU. One axis requires 2 high-speed digital outputs for controlling.			
Digital outputs	Input voltage	Type	Article No.
2	5 V DC	SB 1223	**6ES7223-3AD30-0XB0**
	24 V DC		**6ES7223-3BD30-0XB0**
4	5 V DC	SB 1222	**6ES7222-1AD30-0XB0**
	24 V DC		**6ES7222-1BD30-0XB0**

Recommended controller for SINAMICS S200 PROFINET (PN) version

Version	Integrated interfaces			CPU Processing times for bit operations	Max. number of axes	Work memory	CPU	
	PROFINET IO IRT	PROFINET	PROFIBUS DP					Article No.
SIMATIC S7-1200 Basic Controller for SINAMICS S200 PROFINET (PN) version								
Standard CPUs	1 × PN IO	–	–	85 ns	2	50 KB	CPU 1211C DC/DC/DC	**6ES7211-1AE40-0XB0**
		–	–	85 ns	2	75 KB	CPU 1212C DC/DC/DC	**6ES7212-1AE40-0XB0**
		–	–	85 ns	2	100 KB	CPU 1214C DC/DC/DC	**6ES7214-1AG40-0XB0**
	1 × PN IO	–	–	85 ns	2	125 KB	CPU 1215C DC/DC/DC	**6ES7215-1AG40-0XB0**
	(2-port switch)	–	–	85 ns	2	150 KB	CPU 1217C DC/DC/DC	**6ES7217-1AG40-0XB0**
SIMATIC S7-1500 Advanced Controller for SINAMICS S200 PROFINET (PN) version								
Standard CPUs	1 × PN IO IRT	–	–	60 ns	10	150 KB	CPU 1511-1 PN	**6ES7511-1AK02-0AB0**
	(2-port switch)	–	–	40 ns	10	300 KB	CPU 1513-1 PN	**6ES7513-1AL02-0AB0**
		1 × PN	–	30 ns	30	500 KB	CPU 1515-2 PN	**6ES7515-2AM02-0AB0**
		1 × PN	1 × DP	10 ns	30	1 MB	CPU 1516-3 PN/DP	**6ES7516-3AN02-0AB0**
		1 × PN	1 × DP	2 ns	96	2 MB	CPU 1517-3 PN/DP	**6ES7517-3AP00-0AB0**
		2 × PN	1 × DP	1 ns	128	4 MB	CPU 1518-4 PN/DP	**6ES7518-4AP00-0AB0**
Compact CPUs	1 × PN IO IRT	–	–	60 ns	10	175 KB	CPU 1511C-1 PN	**6ES7511-1CK01-0AB0**
	(2-port switch)	–	–	48 ns	10	250 KB	CPU 1512C-1 PN	**6ES7512-1CK01-0AB0**
Technology CPUs	1 × PN IO IRT	–	–	60 ns	10	225 KB	CPU 1511T-1 PN	**6ES7511-1TK01-0AB0**
	(2-port switch)	1 × PN	–	30 ns	30	750 KB	CPU 1515T-2 PN	**6ES7515-2TM01-0AB0**
		1 × PN	1 × DP	10 ns	80	1.5 MB	CPU 1516T-3 PN/DP	**6ES7516-3TN00-0AB0**
		1 × PN	1 × DP	2 ns	128	3 MB	CPU 1517T-3 PN/DP	**6ES7517-3TP00-0AB0**
		2 × PN	1 × DP	1 ns	192	9 MB	CPU 1518T-4 PN/DP	**6ES7518-4TP00-0AB0**
Fail-safe CPUs	1 × PN IO IRT	–	–	60 ns	10	225 KB	CPU 1511F-1 PN	**6ES7511-1FK02-0AB0**
	(2-port switch)	–	–	40 ns	10	450 KB	CPU 1513F-1 PN	**6ES7513-1FL02-0AB0**
		1 × PN	–	30 ns	30	750 KB	CPU 1515F-2 PN	**6ES7515-2FM02-0AB0**
		1 × PN	1 × DP	10 ns	30	1.5 MB	CPU 1516F-3 PN/DP	**6ES7516-3FN02-0AB0**
		1 × PN	1 × DP	2 ns	96	3 MB	CPU 1517F-3 PN/DP	**6ES7517-3FP00-0AB0**
		2 × PN	1 × DP	1 ns	128	6 MB	CPU 1518F-4 PN/DP	**6ES7518-4FP00-0AB0**
Fail-safe Technology CPUs	1 × PN IO IRT	–	–	60 ns	10	225 KB	CPU 1511TF-1 PN	**6ES7511-1UK01-0AB0**
	(2-port switch)	1 × PN	–	30 ns	14	750 KB	CPU 1515TF-2 PN	**6ES7515-2UM01-0AB0**
		1 × PN	1 × DP	10 ns	80	1.5 MB	CPU 1516TF-3 PN/DP	**6ES7516-3UN00-0AB0**
		1 × PN	1 × DP	2 ns	128	3 MB	CPU 1517TF-3 PN/DP	**6ES7517-3UP00-0AB0**
		2 × PN	1 × DP	1 ns	192	9 MB	CPU 1518TF-4 PN/DP	**6ES7518-4UP00-0AB0**

For SINAMICS S200 PROFINET (PN) version, the AC/DC/RLY and DC/DC/RLY versions of SIMATIC S7-1200 are also possible (CPU 1211C, CPU 1212C, CPU 1214C and CPU 1215C).

SINAMICS S200 as a PROFINET I/O device with PROFIdrive supports technology objects and function blocks of SIMATIC S7-1200, SIMATIC S7-1500 and SIMATIC S7-1500 Technology CPU for speed and positioning control.

For further information about SIMATIC controllers please refer to Catalog ST 70 or to web page:
www.siemens.com/simatic-controller